STUDENT'S HANDBOOK

First Year B. Tech Programme

(ME/CL/EE)

Faculty of Technology & Engineering

Chandubhai S. Patel Institute of Technology

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PART A

GENERAL INFORMATION

CHARUSAT LEGENDS & TERMINOLOGY

CHARUSAT legends are the abbreviation and acronym of the terms used at the university. CHARUSAT legends also include some important terms used at the academic life of the University. The legends are being used to simplify and facilitate rapid communication.

Legends

CHARUSAT	CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CSPIT	Chandubhai S Patel Institute of Technology
DEPSTAR	Devang Patel Institute of Advance Technology and Research
RPCP	Ramanbhai Patel College of Pharmacy
PDPIAS	P D Patel Institute of Applied Sciences
CMPICA	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications
I2IM	Indukaka Ipcowala Institute of Management
ARIP	Ashok & Rita Patel Institute of Physiotherapy
MTIN	Manikaka Topawala Institute of Nursing
CIPS	Charotar Institute of Paramedical Science
CSMCRI	Central Salt And Marine Chemicals Research Institute
CHRF	Charusat Healthcare & Research Foundation
CSRTC	Charusat Space Research And Technology Centre
HRDC	Pri. B. I. Patel Human Resource Development Centre
KRADLE	Dr. K C Patel Research & Development Centre
CREDP	Charusat Rural Education Development Program
UIIC	University Industry Interaction Cell
CDPC	Career Development And Placement Cell
EDIC	Entrepreneurship Development & Incubation Cell
EOC	Equal Opportunity Cell
IQAC	Internal Quality Assurance Cell
CPSH	Cell For Prevention Of Sexual Harassment
ARC	Anti-Ragging Committee
ISC	International Student Cell
GRC	Grievance Redressal Cell
WDC	Women Development Cell
WINCELL	Wireless Information And Networking Cell
CAA	Charusat Alumni Association
ICC	International Center for Cosmology

Terminology

Definitions of Key Words:

1) **Academic Year:** Bachelor of Technology (B.Tech) is the 4-year Course shall be divided into 8 independent semesters with two semesters (One Odd + One Even) in one academic year.
2) **Semester: S**hall constitute of 26 weeks. Each semester shall have minimum 90 days of direct class room teaching, tutorials, counseling, project work and self-learning.
3) **Programme:** An educational programme leading to award of a Degree, Diploma or Certificate.
4) **Course:** Course is a subject in a given semester of a particular programme with given credits and teaching plan leading to an examination.
5) **Choice Based Credit System (CBCS):** The CBCS provides choice for students to select from the prescribed courses (core, elective or minor or soft skill courses).
6) **Credit Based Semester System (CBSS):** Under the CBSS, the requirement for awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.
7) **Credit:** means a unit by which the coursework is measured. As a general guideline, one credit means one hour of class room teaching or minimum one and half to two hours of practical work per week.
8) **Grade Point:** It is a numerical weight allotted to each letter grade on a 10-point scale.
9) **Credit Point:** It is the product of grade point and number of credits for a course.
10) **Letter Grade:** Is a parameter to indicate the performance of a student in a particular course.
11) **Percentage:** The result obtained by multiplying a quantity by a percent. Or proportion or rate per hundred parts. The percent value is computed by multiplying the numeric value of the ratio by 100.
12) **Semester Grade Point Average (SGPA):** It refers to the performance of a student in a given semester. SGPA is ratio of the 'sum of all the products of credit points and grade point earned by the student in all courses of the semester' and the 'total number of credits of all subjects offered in that semester'.
13) **Cumulative Grade Point Average (CGPA):** It refers to the performance of the student in all completed semesters and is equal to Cumulative Grade Point Weighted Average.
14) **Transcript:** A transcript issued to the student at the time of leaving the university will contain a consolidated record of all the courses taken by him / her, grades obtained and the final CGPA.

STUDENT'S CORE COMMITMENTS

CHARUSAT is committed to nurture academic, personal and social values in the students; and expects the students to practice the following Core Commitments of academic, personal and social Responsibility.

ACHIEVING EXCELLENCE

Developing a strong work ethic and consciously doing one's excellent in all aspects of the university.

CULTIVATING, PERSONAL & ACADEMIC INTEGRITY

Recognizing and acting on a sense of respect and dignity both by being honest in relationships and by upholding academic integrity.

CONTRIBUTING TO A LARGER COMMUNITY

Recognizing and acting on one's responsibility to the educational community, the local community, and the wider national and global society.

TAKING SERIOUSLY THE VIEWPOINT OF OTHERS

Recognizing and acting on the responsibility to inform one's own judgment; and engaging various and competing viewpoint as a resource for learning, citizenship, and work.

ENLIGHTENING, ETHICAL AND MORAL REASONING

Developing moral reasoning in ways that incorporate the other four responsibilities and using such reasoning in learning and in life.

ACADEMIC LIFE

CHARUSAT wants all students to achieve their highest academic potential and makes faculty and academic support resources available to assist each student in meeting his/her academic goals.

☛ Teacher

Students needing assistance with a specific course should first seek the help of the teacher. Maintaining continued contact with a teacher and staying informed of academic status in a course is recommended.

☛ Counselor

Each student has a counselor who is knowledgeable about the course the student is pursuing and available to help the student to any academic issue. In addition, the counselor is available to counsel students on all matters related to being a university student as well as on life issues.

☛ Libraries, Laboratories and Workshops

The university has established libraries, laboratories and workshops for an interactive and engaging learning experience.

☛ Attendance Policy

Every institute of CHARUSAT has its own attendance policy; students are required to fulfill the criteria of attendance. Students are required to understand and follow the attendance policy of their institute.

☛ Training and Placement Services

Training and Placement Services at CHARUSAT offers counseling on the choice of a course based on a student's abilities and career interests, as well as networking opportunities with recruiters for potential employment.

ACADEMIC INTEGRITY

CHARUSAT strongly recommends honesty and integrity in all academic work.

Academic Integrity is an ethical practice that means students are achieving academic success fairly. It suggests that all results that are achieved are earned honestly.

Your education is an investment; not maintaining academic integrity may devalue your education, which affects the worth of your degree. Academic Integrity is essential for any society, as people citizens need to trust that who are in positions of authority have earned their credentials rightfully.

Students are expected to exhibit integrity by being truthful about their own academic work and properly acknowledging sources of ideas and information.

➢ Cheating in any form is not tolerated

Cheating Includes:

- Assignment, such as requesting or accepting answers on a quiz or test from another student who has already taken it,
- Discussing test information to any extent with other students, transmitting quizzes or tests or answers to quizzes or tests electronically to other students via cellphone, email, etc.
- Including turning in someone's work as one's own(another student's, a purchased paper from an online source, etc.)

➢ Plagiarism is another form of cheating and academic dishonesty. Intentional or unintentional plagiarism is an offense

Plagiarism includes:

- Use to any degree of the ideas or words of one's source material without proper acknowledgement. Plagiarism typically takes two forms:
- Failure to acknowledge the use of an author's ideas or organization by footnote or identification of the source in the text of the paper.
- Incomplete paraphrase (mere rearrangement of syntax and substitution of synonyms for the author's words) is plagiarism.
- Failure to acknowledge the use of an author's words by quotation marks, as well as by footnote or identification in the text.
- You may consult your teacher or counselor to know more on how to avoid cheating, plagiarism and maintain academic integrity.

SOCIAL LIFE

CHARUSAT's overarching goal is to teach students how to live. To help students experience long-term social success.

CHARUSAT provides ample opportunities to students to enhance their social life on campus. There are many activities, clubs and events organized to enhance student's social interactions and skills.

➢ CHARUSAT PROMOTES

- Healthy Friendship
- Group Activities
- Enhancement of Fraternity
- Respect for Others
- Dignity in Behavior
- Upholding Social and Moral Values
- Outreach Activities in the Society
- Development of Network

CODE OF CONDUCT

Statement of Expectations

As members of the university's community, all students, groups of students, and student organizations are expected to exemplify CHARUSAT's community principles and values, to engage in socially responsible behavior, and to model exceptional conduct, character, and citizenship on campus and beyond.

Parent or Family Contact

Contact with a student's parents or legal guardians may occur or be required in certain circumstances in connection with a matter involving alleged student misconduct or any other academic or personal matter.

Hostel Life

Students are required to follow the rules of the respective residential facility. A decent decorum should be maintained while living in the hostels or any other residential facilities.

Wi-Fi & Internet

CHARUSAT campus is Wi-Fi enabled; moreover, all computers are equipped with internet facility. Students are required to use this facility with maximum integrity. Any misuse of it or misconduct through it will lead to punishment or penalty.

Infrastructure and Instruments

CHARUSAT campus is beautifully designed. All classrooms, laboratories and other areas of the campus are equipped with various amenities and academic instruments. Students are required to use amenities and academic instruments with maximum integrity. Any misuse of it or misconduct through it will lead to punishment or penalty.

Social Media

Social Medias such as Facebook, Twitter, What's App etc. are part of our daily life but it is recommended that all students maintain dignity in the content of posting/ commenting about others and the university.

Communication Devices

Use of cell-phones and other communication devices in the classroom, laboratories, libraries, and at other academic area are prohibited.

Dressing

Students are required to maintain dignified appearance. You may dress up with formal or semi-formal clothes and accessories.

Prayer

CHARUSAT respects all religion. There is a tradition of prayer recitation at the campus premises through Public Address System at 8:55 am. All are requested to maintain the dignity of the prayer time.

DO'S & DON'TS

Do's

- ☑ Set your academic goal high
- ☑ Attend classes regularly
- ☑ Participate in all activities & events
- ☑ Take class notes regularly and refer to them when required
- ☑ Speak to your teachers and counselor about your any academic or personal issues
- ☑ Speak to administration for any issues or problems related to student services
- ☑ Participate in, or create a study group
- ☑ Keep the campus clean
- ☑ Socialize with your peers and develop strong professional relationships
- ☑ Maintain regular contact with your parents to report both good news and bad news
- ☑ Maintain codes of conducts in any kind of communication - oral or written
- ☑ Complete your assignments, projects or any other academic work on time
- ☑ Inspect properly the place before renting resident outside the campus
- ☑ Ask what you can do to help others
- ☑ Consult CHARUSAT website and notice boards regularly for any updates and announcements

Don'ts

- ☒ Wander around the unknown peripheral areas of the campus
- ☒ Share personal information to unknown
- ☒ Damage any property of the campus
- ☒ Leave your personal belongings unattended
- ☒ Participate in or initiate gossips or rumors
- ☒ Make loud noise or create confusion in the class room, auditorium or elsewhere in the building
- ☒ Use abusive language
- ☒ Assume your first and second semester marks don't count. CGPA's of your whole program are looked at during applications for further studies and career
- ☒ Use cell-phones or any other communication devices in the classroom or any other part of the building where academic activities are going on

STUDENT SERVICES

CHARUSAT strongly believes that a student's life at the campus should be comfortable and hassle-free, and for that the university has carefully designed various services for the students. You are requested to avail the services as and when required.

🕮 Library 🕮

- ☞ The Knowledge Resource center (Central Library) - a proud partner in the institute's march towards its vision, plays a vital role in acquisition, organization and dissemination of knowledge. You shall need this for almost all your academic assignments!
- ☞ It has an excellent collection of both print and electronic books, journals, technical reports, back volumes and other reading material. It has adequate infrastructure to meet its requirements, has computerized all its operation using software developed in-house, and provide access to the collection through Online Public Access Catalogue (OPAC).
- ☞ Along with the Central Library, there are Institute Level Libraries in each Institute Building. The Libraries are enriched with more than 50,000 books and 15,000 journals (including e-journals).
- ☞ The Knowledge Resource Centre maintains a e-resource access center containing 25 computer terminals for the students in which they can access national and international e resources namely IEEE, ASME, AIP, IOP, IPS, CSPIT library database containing CD's, e-books, journals, Project Reports, Syllabus, University Exam papers through Intranet (ftp://172.16.1.14). Moreover separate computer terminals provide to students with CD writer and USB port for their presentation of seminar, project work and day to day work. The E-resources can be accessed through other computer terminals anywhere in campus. Try learning more on this!!

For any queries, you may please contact the Library office
Contact Person: Mr. Dinesh Patel, Librarian
Contact Number: (02697) 265032; 9909543820
Contact Email: dineshpatel.lib@charusat.ac.in

🏠 Residences 🏠

- The residences for girls are available at CHARUSAT Campus and residences for boys are available adjacent to campus.
- The life in hostels enables students to spend ample time at the university utilizing library and other facilities to ensure they develop academically and acquire the necessary skills that can be obtained only through experience.

For any queries, you may please contact
Contact Person: Mr. Hasmukh Patel
Contact Number: (02697) 265004; 9426500630
Contact Email: hasmukhpatel.adm@charusat.ac.in

🚍 Transport Services 🚍

- CHARUSAT has outsourced bus services for providing the transportation facilities to the students.
- A fleet of buses are there for transporting students and staff from different locations in Ahmedabad, Vadodara, Anand and Nadiad and nearby villages every day. A VITCOS bus service has been initiated for students from Anand at a very minimal rate.
- Students are supposed to pay directly to the travel company either monthly or six-monthly or yearly installments.

For any queries, you may please contact
Contact Person: Contact Number: (02697) 265018;

🚑 Healthcare 🚑

- CHARUSAT Hospital is established to provide primary health care services for emergency and daily health cases. It organizes periodical health screening programs and health awareness activities and campaigns.

For any queries and emergencies you may please contact
CHARUSAT Hospital, Reception Desk
Contact Number: (02697) 265291; 9537927873

ⓘ Student Safety Cells ⓘ

- CHARUSAT believes that it should be a safe workplace as well as a safe place to study. All the students of CHARUSAT may avail following help if need arises:

Cell for Prevention of Sexual Harassment (CPSH)
Please call on 7600414303 for complaints
https://www.charusat.ac.in/ac/cells-at-charusat/
Anti-Ragging Committee (ARC)
Please call on 09925830781 for complaints.
https://www.charusat.ac.in/ac/cells-at-charusat/

💻 Wincell 💻

- The Wireless Information and Networking Cell, is the Cell looking after IT Infrastructure of CHARUSAT. CHARUSAT is a Wi-Fi zone with 100 mbps connectivity. Internet is available on each computer terminals. For all your queries like Internet Access, Printing or other such issues, contact Wincell Department.

For Wincell Department, you may please contact
Contact Person: Mr. Ritesh Bhatt
Contact Number: (02697) 265106; 9924444809
Contact Email: riteshbhatt.win@charusat.ac.in

E-Governance

- Almost all the process of CHARUSAT are computerized and connected through customized Entrepreneurs Resource Planning Software. This whole system is called E-Governance. The students shall be needing to access these system for registration, syllabus, time-table, attendance, student I-card, Fees and other receipts, exam results, convocation form, interaction platforms with teachers like blogs, etc.
- Each department has an E-Governance Representative. You may contact Principal of your Institute for further details.

⌘ Study Foyer ⌘

- There is a special area dedicated for reading. It is located on the first floor near central library.

✏ Reprography & Stationery ✏

- There are facilities for reprography (photocopy) and buying stationeries on campus. It is located on the first floor near central library.

💰 Bank, Post & ATM 💰

- There are facilities of banking and post in campus. It is located in the central administrative building, and ATM facility is just near to campus.

CAREER DEVELOPMENT & PLACEMENT CELL

The Motive of Charotar University of Science and Technology (CHARUSAT), Changa is to help society to develop towards a better future. We believe in providing value based education to the students so that they can be better employable candidates and more importantly an individual contributing to the organization and the society as a whole.
For the same purpose, a dedicated centralized Career Development and Placement Cell has been constituted on the campus. The Cell coordinates all the Training and Placement activities of different institutes of the University and enhances Industry Institute Interaction

Training Activities

- Training activities are arranged at two levels i.e. CDPC and institute or department wise. The training programmes are concentrated towards providing students with ample exposure to recruitment patterns and skill requirements of different private and public sector organizations. The training programs offered include Behavioral Skills, Technical Skills, Personality Improvement, and Communication Skills.

- Each Institute provides Training in accordance with the curriculum/course. Training is provided to them in coordination with different Private/Public organization according to prevalent Industry demand. This is undertaken to provide the student with the real time environment in industry so that the student can have a firsthand practical experience of the latest practices and technologies. The duration of the training can differ according to specific course and its need. The Training Cell also arranges program on Behavioral Skills and Technical Skills Training to the students prior to facing Campus Placements.

Placement Activities

- The Robust and Dedicated Centralized Placement Cell facilitates On-Campus / Pooled Campus / Off-Campus activities to provide job assistance to students in leading organization. The Placement Department in coordination with Institute Placement Coordinator invites reputed organization for placements activities. All the Major Industry and Sector are targeted to make provide ample opportunities to the students.

Career Guidance

- Career Guidance Career Development and Placement Cell also organizes Seminars/Workshops/Training Programs/Guest Lectures on various career avenues and options that a student could explore (GATE/GRE/TOEFL/CAT/UPSC/GPSC etc). Sectoral Inputs are provided to students to make wise choices about the sector they choose to build their career.

🏛 Resume Building and Interview Preparation 🏛

☞ The cell guides students on how to prepare appropriate Resume/CV including video resume and how to prepare for the interviews. Special sessions on technical / aptitude / soft and employability skills are conducted in house as we well as by inviting Industry experts to provide students more exposure and improve their employability.

🕴 Placement Infrastructure 🕴

☞ The University has state of the art infrastructure with 24X7 Wi-Fi Campus, Internet Connectivity, Several Large network line Labs and Auditoriums and Seminar halls for conducting Placement Drives. CHARUSAT has also hosted Pooled Campus Drives for Infosys, Amdocs, and Alembic to name a few, for the entire region.

For further information, you may please contact
Training & Placement Officer

Contact: Prof. Ashwin Makawana,

Mr. Divyang Purohit, Mr. Ernest Stevens

Contact Number: (02697) 265213; 9913686259

Email: tnp@charusat.ac.in, divyangpurohit.tnp@charusat.ac.in

There are also T&P co-coordinators in each Department; you may contact HOD of your department to know more.

STUDENT PROFESSIONAL ACTIVITIES

✡ Student Chapters, Societies and Academies ✡

	IEEE is the world's largest professional association dedicated to advancing technological innovation and excellence for the benefit of humanity. CHARUSAT is an official Student Branch of IEEE. Tech Enthusiastic Student IEEE Members of CHARUSAT organizes technical activities like workshops, seminars and Technical Festivals to motivate and increase other student's interest in technical research and innovations. IEEE Student Branches are established at universities and colleges around the world. Within IEEE, activities are organized geographically by Region and local Section. Student Branches in R10 with Counsellor & Chair contact August 2015.
IIT Certification Through NPTEL Local Chapter	NPTEL has been offering online certification for its courses, the highlight being the certification exam through which the student gets an opportunity to earn a certificate form the IITs! To take this initiative forward and to encourage more students across colleges to participate in this initiative, CHARUSAT had started NPTEL Local chapter during December 2015 which is currently coordinated by CSPIT.
	Computer Society of India is the first and largest body of computer professionals in India. It was started on 6 March 1965 by a few computer professionals and has now grown to be the national body representing computer professionals. The Computer Society of India is a non-profit professional meet to exchange views and information learn and share ideas. The wide spectrum of members is committed to the advancement of theory and practice of Computer Engineering and Technology Systems, Science and Engineering, Information Processing and related Arts and Sciences.
	The Association for Computing Machinery is an international learned society for computing, founded in 1947. ACM is the world's largest educational and scientific computing society, delivers resources that advance computing as a science and a profession. ACM is solely dedicated to computing. ACM provides the computing field's premier digital library and serves its members and the computing profession with leading-edge publications, conferences and career resources. CHARUSAT is an official Student Chapter of ACM initiated by Department of Information Technology, CSPIT on 26 th August 2016.
	Microsoft Technology Associate (MTA) Microsoft Certified Solutions Associate (MCSA) Microsoft Certified Solutions Expert (MCSE) Microsoft Certified Solutions Developer (MCSD)
	Oracle Certified Associate (OCA) Oracle Certified Professional (OCP) Oracle Certified Master (OCM) Oracle Certified Expert (OCE) Oracle Certified Specialist (OCS) Oracle Certified Associate Java SE Programmer. Oracle Certified Professional Java SE Programmer.
	Cisco Certified Network Associate 1 (CCNA 1) Cisco Certified Network Associate 2 (CCNA 2) Cisco Certified Network Associate 3 (CCNA 3) Cisco Certified Network Associate 4 (CCNA 4) Cisco Certified Network Professional (CCNP)

	The chapter is established in the year 2007 to provide a platform to students to help in applying their technical skills to practical aspects. This chapter has conducted various expert lectures, Two National level TECHFEST in the year 2008 and 2010 and more than 15 various workshops to practical exposure. Some of the Chapter activities are: Technical Quizzes, Circuit Designing Circuit analysis and fault finding, Elocution, Seminars and workshops, Poster presentations, Concept and idea presentation and exhibitions, Technical treasure hunts, Expert lectures
	The ISHRAE student chapter in CHARUSAT have been established to promote the activities to protect the Environment, improve Indoor Air Quality, help Energy Conservation, and provide continuing education to the Members and others in the HVAC & related user Industries and offer certification programs, career guidance to students. Some of the Chapter activities are: Industrial visit, Guest Lecture, Quiz, Seminars etc.
	SAEINDIA is an affiliate society of SAE International, registered as an Indian non-profit engineering and scientific society dedicated to the advancement of mobility community in India. As an individual member driven society of mobility practitioners, SAEINDIA comprises members who are individuals from the mobility community, which includes engineers, executives from industry, government officials, academics and students. Principal emphasis is placed on transport industries such as automotive, aerospace, and commercial vehicles. SAEINDIA sections were formed all across the country. In 2008, Formula SAEINDIA was launched with the name of SUPRA SAEINDIA. This event provides a real world engineering challenge for the SAEINDIA student members that reflects the steps involved in the entire process from design and engineering to production to marketing and endurance. Since last four years 'Team Ojaswat' a team of 25 students from Charusat University has been participating in SUPRA SAEINDIA. Right from SUPRA'14 - secured 3rd rank, SUPRA'16- secured 5^{th} rank all over India. And the team cleared technical Inspection in First attempt as a debut in Formula Bharat an International Event in Jan'17.
	Society of Civil Engineering (SCE) is a non-registered non-profit academic initiative taken by the Department of Civil Engineering, Chandubhai S. Patel Institute of Technology under the governance of Charotar University of Science and Technology. The SCE is a society with a group of nascent engineers committed to experience high applications of civil engineering concepts on field with an objective of promoting civil engineering, bring new technologies in civil engineering to the grass root level of the human society through students and promote environmentally sustainable construction technologies.
FESTO Centre of Excellence 	FESTO Didactic has been recognized worldwide for the development of high-quality, intuitive learning systems for technical education. FESTO Didactic brings over 40 years of experience into developing solutions for fast learning and successful retention over a broad spectrum of technologies. FESTO Centre of Excellence at CHARUSAT has learning and training facilities for pneumatic systems, hydraulic systems and factory automation. It is an experience center in true sense.
International Center for Cosmology ICC	The International Center for Cosmology has been initiated with the purpose of conducting frontier level research on the nature and structure of our universe. There have been path-breaking developments on our knowledge of cosmos in the past decades, and exciting new horizons are emerging. The International Center for Cosmology which was inaugurated in 2018, plans to contribute in a big way in these exciting developments, thus fostering a vibrant research culture within a University environment.

STUDENT ACTIVITIES & EVENTS

🏳 NSS & NCC 🏳

- CHARUSAT offers training for NSS and NCC for interested students. Interested students may please contact the Sports and Gymnasium Coordinator

For any queries or details, you may please contact
Contact Person: Mr. Yogesh Jani
Contact Number: (02697) 265036; 9558295583
Contact Email: yogeshjani.sports@charusat.ac.in

🏆 Co-Curricular & Cultural Events 🏆

- A vast range of cultural and social activities are available to CHARUSAT students, faculty and staff. Getting involved in campus life is the quickest way to become a part of the University community, and to create one's own CHARUSAT experience. Campus life activities are built around the concepts of encouraging each community member to express his or her talents and to respect all members of our pluralistic community.
- The students can exhibit their special talents by the multiple college and inter-college competitions within and outside the campus. CHARUSAT organizes a four day gala event of University level Cultural Competition named SPOURAL. In addition, faculty of Technology & engineering (CSPIT & DEPSTAR) organizes COGNIZANCE - TECH FEST, a state level technical event annually.
- The University also encourages the students and staff to celebrate all the varied festivals at the campus like Uttrayan, Holi, Navratri, Ganesh Chaturthi, etc.

RECREATION & EFRESHMENTS

🏋 Sports & Gymnasium 🏋

- ☞ CHARUSAT campus offers wide range of team sports, exercises, fitness and other related activities. Selected activities include various indoor and outdoor games like badminton, table tennis, cricket, volleyball and football, gymnasium, etc.
- ☞ The gym is open from 9am to 5pm. Please remember to bring your student ID card every time you visit the gym and you also have to bring sport shoes. Non-sport shoes will not be allowed in the gym.

Sr. No.	Activities	Timing
1.	University Fitness Center Ground Floor, ARIP Building	Morning 6:00 a.m. to 8:00 a.m. For Girls and Female Faculty
		Evening 4:30 p.m. to 7:30 p.m. For Boys and Male Faculty
2.	University Gymnasium 1st Floor, Hari Om Food Plaza Building	Morning 6:00 a.m. to 8:00 a.m. for Boys and Male Faculty
		Evening 4:30 p.m. to 7:00 p.m. For Girls and Female Faculty
3.	Indoor Sports for Boys and Girls	Evening 4:30 p.m. to 7:00 p.m.

You may please contact the Sports and Gymnasium

Coordinator

Contact Person: Mr. Yogesh Jani
Contact Number: (02697) 265036; 9558295583
Contact Email: yogeshjani.sports@charusat.ac.in

🍽 Food 🍽

- ☞ The campus cafeteria and other food outlets are open every class day, serving breakfast, lunch and snacks. You can bring your own snack/lunch also. The Campus Cafeteria is situated at Lake Side serving multiple cuisine food. The other Fast Food Outlets like Nescafe and Iceberg are also available at the campus.

For any queries, you may please contact
Contact Person: Mr. Hasmukh Patel
Contact Number: (02697) 265004; 9426500630
Contact Email: hasmukhpatel.adm@charusat.ac.in

FINANCIAL INFORMATION

● Insurance ●

CHARUSAT believes in the safety of the students. Hence, it has insured each and every student of the campus with the Group Personal Accident Insurance Policy.

For Further Details, You May Please Contact

Contact Person: Mr. Bhavdip Patel, Chief Accounts Officer

Contact Number: (02697) 265024; 9925946858

Contact Email: bhavdippatel.acc@charusatac.in

● Financial (Aid) Scholarships ●

CHARUSAT provides its students' with a number of financial support opportunities. These opportunities are exclusively focused on providing support for students whose financial conditions may prevent them from continuing their education.

➢ **Following Scholarship schemes are available which are mentioned below**

Name of Scholarship	Beneficiaries
CHARUSAT Merit Scholarship (CMS-2015) (CMS-2016)	All students who satisfy stipulated percentile criteria.
GPAT Scholarship (By AICTE)	As per Government norms.
Government Scholarship	All students of SC, ST and SEBC category, Free ship card for SC students, Chief Minister Scholarship Scheme, Mukhyamantri Yuva Swavalamban Yojna (MYSY (http://mysy.guj.nic.in/).
MOMA Scholarship	Students belonging to minority communities.
Late Maniben Shankarbhai Patel Scholarship (Implemented from 2014-15)	1st Rank of 2nd, 3rd & 4th year student of B.Sc Nursing Program.
Late Shankarbhai Chhaganbhai Patel Scholarship (Implemented from 2014-15)	1st Rank of 2nd, 3rd & 4th year student of B.Pharm Program.
Late Dahiben Ravjibhai Patel & Dineshbhai Ravjibhai Patel Merit Cum Means Scholarship (2016-17)	Meritorious & Economically Constrained Students of IT branch of CSPIT
Urmil & Mayuri Desai Family Trust Scholarship (2016-17)	Meritorious & Economically Constrained Students of Engineering of CSPIT

Name of Scholarship	Beneficiaries
Umedbhai Dharamdas Patel (Nar) Charitable Trust Scholarship	Meritorious & Economically Constrained Students of Selected Course
CHAMOS CHARUSAT Merit Cum Means Scholarship	Students of Charotar Moti Satavis Patidar Samaj, who take admission at CHARUSAT University.

For any further queries, you may please contact

Contact Person: Mr. Darshan Patel

Contact Number: (02697) 265027; 8866286880

Contact Email: darshanpatel.adm@charusat.ac.in

PART B

ACADEMIC INFORMATION

Faculty of Technology & Engineering

Chandubhai S. Patel Institute of Technology
(CSPIT)

ACADEMIC REGULATIONS
&
SYLLABUS

(Choice Based Credit System)

Bachelor of Technology Programme

(First Year B. Tech Programme ME/CL/EE)

ACADEMIC REGULATIONS

Bachelor of Technology (ME/CL/EE) Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)

CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand

Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in

www.charusat.ac.in

Year – 2019-2020

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1) System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2) Duration of Programme

i)	**Undergraduate programme**	**(B.Tech)**
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3) Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4) Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5) Programme structure and Credits

As per annexure – 1 attached

6) Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7) Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.

7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations are as follows.

Minimum percentage marks to be obtained in end-semester University Examination (for applicable course)	Minimum Overall percentage marks to be obtained in each course.
40%	45%

7.2.2 If a candidate obtains minimum required percentage of marks in end-semester university examination in applicable course but fails to obtain minimum required overall percentage of marks, he/she has to repeat the examination till the minimum required overall percentage of marks are obtained.

8) Grade Point System

1. **The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:**

Table 1 **Grade Point System (UG)**

Range of Marks (%)	**≥80**	**<80 ≥73**	**<73 ≥66**	**<66 ≥60**	**<60 ≥55**	**<55 ≥50**	**<50 ≥45**	**<45**
Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

2. **The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:**

(i) SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

9) Award of Class

☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	CGPA ≥ 7.5 & ≤ 10.0
First class	CGPA ≥ 6.0 & < 7.5
Second Class	CGPA ≥ 5.0 & < 6.0
Pass Class	CGPA < 5.0

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9. 1. Maximum duration allowed for Completion of a programme

☞ Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10) Detention Criteria

☞ No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

☞ A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

☞ A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

11) Transcript

☞ A transcript issued to the student at the time of leaving the university will contain a consolidated record of all the courses taken by him / her, grades obtained and the final CGPA.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING

(FTE)

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 4 types of courses: Foundation courses, Core courses, Elective courses and Non-credit (audit) courses.

1.1) Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2) Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. **University Core courses(UC)**:

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. **Programme Core courses(PC)**:

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3) Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

D. Cluster Elective Course (CE):

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.4) Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be consider to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5) Medium of Instruction

The Medium of Instruction will be English.

Annexure – 1

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY													
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ME/CL/EE ENGINEERING													
CHOICE BASED CREDIT SYSTEM													
Sem	**Course Code**	**Course Title**	Teaching Scheme					Examination Scheme					
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
First Year Sem I	MA143	Engineering Mathematics - I	4	0	1	5	4	30	70	0	0	100
	ME146	Engineering Graphics	2	2	1	5	3	30	70	25	25	150
	ME142	Workshop Practices	0	2	0	2	1	0	0	25	25	50
	CL143	Engineering Mechanics	3	2	1	6	4	30	70	25	25	150
	PY141.01	Engineering Physics	3	2	0	5	4	30	70	25	25	150
	HSXXX	A Course Liberal Arts (HS Elective - I)	2		0	2	2	0	0	30	70	100
	Assignment Practices /Student counselling /Remedial classes / Library/ Sports/ Extracurricular &co-curricular					7						
	Total					**32**	**18**					**700**
First Year Sem II	MA144	Engineering Mathematics - II	4	0	1	5	4	30	70	0	0	100
	ME147	Basics of Civil & Mechanical Engineering	3	2	0	5	4	30	70	25	25	150
	CL142.01	Environmental Sciences	0	2	0	2	2	0	0	30	70	100
	EE145	Basics of Electronics & Electrical Engineering	3	2	0	5	4	30	70	25	25	150
	IT143	Fundamentals of Computer Programming	2	2	0	4	3	30	70	25	25	150
	HS 126.01A	Communication Skills - I	2		0	2	2	30	70	0	0	100
	Assignment Practices /Student counselling /Remedial classes / Library/ Sports/ Extracurricular &co-curricular					9						
	Total					**32**	**19**					**750**

B. Tech. (ME/CL/EE)
Programme

SYLLABI
(Semester – 1)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Table of Contents

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA143: Engineering Mathematics-I

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Total	Credit
Hours/week	4	1	5	4
Marks	100	-	100	

A. Outline of the course:

Sr No.	Title of the unit	Number of hours
1	Higher order derivatives and applications	16
2	Complex numbers and Roots of polynomial Equations	14
3	Matrix Algebra- I	12
4	Partial differentiations	08
5	Applications of Partial differentiations	10
	Total hours	60

Total hours (Theory): 60 Hrs.

Practical Hours: 00 Hrs.

Total hours: 60 Hrs.

B. Detailed Syllabus:

1 Higher order derivatives and applications: 16 Hours 27%

1.1 Set theory and Function

1.2 Limit, Continuity, Differentiability for function of single variable and its uses. Mean Value Theorem, Local Maxima and Minima

1.3 Successive differentiation: n^{th} derivative of elementary functions: rational, logarithmic, trigonometric, exponential and hyperbolic etc.

1.4 Leibnitz rule for the n^{th} order derivatives of product of two functions

1.5 Tests of convergence of series viz., comparison test, ratio test, root test, Leibnitz test. Power seriesexpansion of a function: Maclaurin’s and Taylor’s series expansion.

1.6	L'Hospital's rule and related applications, Indeterminate forms		
2	**Complex numbers and Roots of polynomialEquations:**	**14 Hours**	**23%**
2.1	Complex numbers and their geometric representation		
2.2	Complex numbers in polar and exponential forms		
2.3	De Moivre'stheorem and its applications		
2.4	Exponential, Logarithmic, Trigonometric and hyperbolic functions.		
2.5	Statement of fundamental theorem of Algebra, Analytical solution of cubic equation by Cardan's method		
2.6	Analytic solution of Biquadratic equations by Ferrari's method with their applications.		
3.	**Matrix Algebra- I:**	**12 Hours**	**20%**
3.1	Definition of Matrix, types of matrices and their properties		
3.2	Determinant and their properties		
3.3	Rank and nullity of a matrix		
3.4	Determination of rank		
3.5	The inverse of a matrix by Gauss Jordan method.		
3.6	Solution of a system of linear equations by Gauss elimination and Gauss Jordan Methods.		
4.	**Partial differentiations:**	**08 Hours**	**13%**
4.1	Partial derivative and geometrical interpretation		
4.2	Euler's theorem with corollaries and their applications		
4.3	Chain rule		
4.4	Implicit functions		
4.5	Total differentials		
5.	**Applications of Partial differentiations:**	**10 Hours**	**10%**
5.1	Maclaurin's and Taylor's series expansion in two variables		
5.2	Tangent plane and normal line to a surface		
5.3	Maxima and Minima		
5.4	Langrage's method of multiplier		
5.5	Jacobian		
5.6	Errors and approximations		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quiz (surprise test) /Oral tests/ Viva/Assignment/Tutorials will be conducted which carries 10% component of the overall evaluation.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	develop skill of successive differentiation, utilize appropriate theory and computational techniques to construct Taylor's series with its interval of convergence for using in a variety of applications such as approximating values, creating series representation and behaviour of a functions, use L'Hospital's rule to compute limits of the indeterminate forms.
CO2	perform basic mathematical operations with complex numbers in Cartesian and polar forms, know methods of finding the n^{th} roots of a complex number and solutions of simple polynomial equations, work with functions of complex variable.
CO3	find determinant and inverse of a square matrix, evaluate rank and nullity of a matrix, solve system of linear equations by using concept of matrices which are useful in various fields of engineering.
CO4	evaluate partial derivatives including higher order derivatives, solve problems using the chain rules, Euler's theorem with corollaries, implicit function and total differentials.
CO5	expand any function of two variables in ascending power of variables, solve problems using the techniques of multivariable calculus in various branches of engineering.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	1	-	2	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	1	-	-	-	-	-	-	-	-	-
CO5	3	2	1	1	1	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below

1: Slight (Low) 2:Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-":

Recommended Study Material:

- **Text Books:**

1. Erwin Kreyszig; Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. H. K. Dass and RajnishVerma; Higher Engineering Mathematics, S Chand & Co Pvt Ltd. 2012.
3. B. S. Grewal; Higher Engineering Mathematics, Khanna Publ., Delhi, 2012

- **Reference Books:**

1. M. D. Weir *et al.;* Thomas' Calculus, 11th Ed., PearsonEducation, 2008.
2. James Stewart; Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. C. R. Wylie and L. C. Barrett; Advanced Engineering Mathematics. 1982., McGraw-Hill Book Company.
4. Michael D. Greenberg; Advanced Engineering Mathematics. Prentice-Hall, 1988.

- **Web Materials:**

1. https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf
2. http://nptel.ac.in/courses/111107108/
3. http://nptel.ac.in/courses/122101003/
4. http://nptel.ac.in/courses/111104085/

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
ME146: Engineering Graphics

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	1	5	**3**
Marks	100	50	0	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Fundamentals of Engineering Graphics	05
2	Projections of Points and Lines	03
3	Projections of Planes	03
4	Projections & Section of Solid	04
5	Orthographic Projection*	05
6	Isometric Projections*	04
7	Computer Graphics*	03
8	Development of Lateral Surfaces	03

Total hours (Theory): 30 Hrs.

Practical Hours: 30 Hrs.

Total hours: 60 Hrs.

B. Detailed Syllabus:

1. Fundamentals of Engineering Drawing **05 Hours 17%**

1.1 Importance of engineering drawing, drawing instruments and materials, BIS and ISO

1.2 Different types of lines used in engineering practice, methods of projections as per SP 46-1988.

1.3 Engineering scale.

1.4 Engineering curve.

2.	**Projections of Points and Lines**	**03 Hours**	**10%**
2.1	Introduction to methods of projections		
2.2	Projections of lines inclined to both the planes		
3.	**Projections of Planes**	**03 Hours**	**10%**
3.1	Projection of plane		
3.2	Auxiliary projection method		
4.	**Projections & Section of Solid**	**04 Hours**	**13%**
4.1	Projection of solids		
4.2	Sectional view		
4.3	True shape of Sections		
4.4	Auxiliary Inclined Plane (AIP), Auxiliary Vertical Plane (AVP)		
5.	**Orthographic Projection**	**05 Hours**	**17%**
5.1	Principle projection		
5.2	Methods of first and third angle projection with examples / problems		
6.	**Isometric Projections**	**04 Hours**	**13%**
6.1	Terminology, Isometric scale		
6.2	Isometric view and Isometric projection with examples / problems		
7.	**Computer Graphics**	**03 Hours**	**10%**
7.1	Introduce computer graphics		
7.2	Importance of computer graphics		
7.3	Demonstration of CAD Modeling software		
7.4	Training of Fusion 360 software		
8.	**Development of Lateral Surfaces**	**03 Hours**	**10%**
8.1	Method of Development		
8.2	Developments of cylinder, cone, prism, pyramid		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.

- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Drawing sheets/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Realize and interpret various engineering drawings.
CO2	Understand the concept and application of engineering scale and engineering curve.
CO3	Visualize three-dimensional drawing of engineering components through orthographic, sectional orthographic and isometric drawing.
CO4	Explore the basic principle of projections and development of lateral surface of solid.
CO5	Use the computer for geometric modeling.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. N. D. Bhatt & V. M. Panchal, "Engineering Drawing", Charotar Publishing House Pvt. Ltd.
2. P. J. Shah, "Engineering Graphics", S. Chand Publishing & Co.

❖ **Reference book:**

1. P.B. Patel & P.D. Patel, "Engineering Graphics", Mahajan Publishing House.
2. Arunoday Kumar, "Engineering Graphics", Tech-Max Publication.
3. Gopal Krishna K.L., "Engineering Drawing", Subhas Publications
4. Venugopal, K., "Engineering Drawing made Easy", Wiley Eastern Ltd.
5. M.L. Agrawal & R.K. Garg, "Engineering Drawing", Vol. I, Dhanpatrai & Co.
6. T.E. French, C.J. Vierck & R. J. Foster, "Graphic Science and Design", McGraw Hill.
7. W. J. Luzadder & J. M. Duff, "Fundamentals of Engineering Drawing", Prentice Hall.
8. K. Venugopal, "Engineering Drawing and Graphics", New Age international Pry. Ltd.

❖ **Web Materials:**

1. http://nptel.ac.in/courses/112103019/
2. http://nptel.ac.in/downloads/112105125/
3. http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101
4. http://nptel.ac.in/courses/105107122/
5. https://law.resource.org/pub/in/bis/S01/is.sp.46.2003.pdf

FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
ME142: WORKSHOP PRACTICES

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**1**
Marks	-	50	-	50	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to workshop facility.	02
2.	Carpentry shop	06
3.	Fitting shop.	08
4.	Different Metal Joining Processes.	04
5.	Smithy shop.	02
6.	Sheet metal working.	02
7.	Plumbing shop.	02
8.	Introduction to machine tools.	02
9.	Injection molding process.	02

Total hours (Lab): 30

Total hours: 30

B. Detailed Syllabus:

1. Introduction to workshop facility **02 hours 07%**

Familiarization with work shop facility, Introduction to different shops of the workshop.

2.	**Carpentry Shop** Introduction to different tools of carpentry shop, Making of drawing of the job to be made, Making of finished job as per drawing out of the given raw material of wood, Identification on the job for traceability.	**06 hours**	**20%**
3.	**Fitting Shop** Introduction to different tools of fitting shop, Making of drawing of the job to be made, Making of finished job as per drawing out of the given raw material. Identification on the job for traceability	**08 hours**	**27%**
4.	**Different Metal Joining Processes** Introduction to different tools of welding shop, Making of drawing of the job to be made, Making or demonstration of finished job as per drawing, Introduction to Soldering and brazing of metal joining process, Joining of two metal sheet or plate by Riveting, Making of drawing of the job to be made by riveting	**02 hours**	**14%**
5.	**Smithy Shop.** Introduction to different tools of smithy shop, Making of drawing of the job to be made for Cold smithy, Making or demonstration of finished job as per drawing, Making of drawing of the job to be made for Hot smithy, Making or demonstration of finished job as per drawing	**04 hours**	**07%**
6.	**Sheet Metal Working.** Introduction to different tools of sheet metal working shop, Making of drawing of the job to be made from sheet metal, Making or demonstration of finished job as per drawing	**02 hours**	**06%**
7.	**Plumbing Shop.** Introduction to all plumbing tools, Demonstration of plumbing on the piping model	**02 hours**	**06%**
8.	**Introduction to Machine Tools.** Detailed introduction to Lathe machine, Shaping machine, Drilling machine, Grinding machine, Milling machine, Bending machine, Mechanical press.	**02 hours**	**07%**
9.	**Injection molding process.**	**02 hours**	**06%**

Introduction and demonstration to Injection Molding Process for making job out of plastic material

C. Instructional Method and Pedagogy:

- ❖ Attendance is compulsory in laboratory session.
- ❖ Journal writing based on above course content and practical work in form of performance practical's by preparing job at the workshop floor.
- ❖ In the laboratory discipline and behavior will be observed strictly.
- ❖ All the students must follow code of conduct during working at the shop floor.
- ❖ Journal should be submitted to the respective course teacher within the given time limit.

Course Outcome (COs):

On the completion of the course one should be able to:

CO1	Recognize essential tools and process of carpentry.
CO2	Understand the joining process like welding, soldering and brazing.
CO3	Recognize and use essential tools and different machines.
CO4	Recognize various forging and forming processes with the aid of smithy process.
CO5	Understand different types of sheet metal joints which are useful in working.
CO6	Recognize process of fitting,different types of fittings plumbing tools and plastic molding processes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	1	-	-	-	-	-	-	-	-	-	-	-	-	-

Recommended Study Material:

- ❖ **Text book:**

1. Anderson, James, and Earl E. Tatro. "Shop Theory. 5th ed." (1968).

2. Bawa, H. S. Workshop Technology. India: Tata McGraw-Hill, (1995).

3. Choudhury, Hajra. "Elements of Workshop Technology, Vol. I & II." Media Promotors Pvt Ltd (2009).

4. Raghuwanshi, B. S. Course in Workshop Technology. Dhanpat Rai and Company (P) Limited, (2009).

❖ **Reference book:**

1. Chapman, W. A. J. "Workshop Technology Part 1-3." (1998).

2. Tejwani V.K., "Basic Machine Shop Practic e Vol. I, II", Tata McGraw Hill Pub. Co., New Delhi, (1989).

3. Arora B.D., "Workshop Technology Vol. I, II", Satya Prakashan, New Delhi, (1981).

❖ **Web Materials:**

1. www.nptel.iitm.ac.in

FACULTY OF TECHNOLOGY & ENGINEERING
M S PATEL DEPARTMENT OF CIVIL ENGINEERING
CL143 ENGINEERING MECHANICS

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100	-	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	04
2	Fundamental of Statics	20
3	Friction	05
4	Truss	05
5	Centroid and Centre of Gravity	06
6	Fundamentals of Kinematics and Kinetics of Particles	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1 Introduction **04 Hours 08%**

Introduction of Mechanics, Fundamental concepts and idealization of mechanics, Fundamental principles & Laws of mechanics, Scalar and Vector Quantities, Components, unit vectors and position vector, Composition and resolution of vector, System of Units

2 Fundamental of Statics **20 Hours 45%**

2.1 Coplanar Concurrent Force system

Introduction of Force, Effect of force and Characteristics of force, Types of force, Type of force systems, Principle of Transmissibility, Resultant of force systems, Resolution of a

	single force, Composition and Resolution of force system, Resolution method for coplanar concurrent force system		
2.2	**Moments and Couples** Moment of a force, Principle of moments, Coplanar applications, Parallel force system, Couples, Equivalent couples, Operations with couples, Equivalent system of forces		
2.3	**Coplanar Non-Concurrent Force system** Introduction, Resultant of coplanar non-concurrent force system, Concentrated and distributed loads		
2.4	**Equilibrium of Rigid bodies** Equilibrium, Resultant & Equilibrant, Principle of action and reaction, Free body diagram & Lami's theorem, Tensions of strings, condition of equilibrium for Coplanar concurrent forces & Coplanar non-concurrent forces, Equilibrium of Coplanar concurrent forces, Equilibrium of Coplanar non-concurrent forces		
3	**Friction** Friction and its applications, Types of friction and Laws of dry friction, Angle of friction, Angle of repose, Coefficient of friction, Block Friction, Ladder friction, Wedge friction	**5 Hours**	**11%**
4	**Truss** Classification of Truss, Perfect & Imperfect truss, Analysis of pin jointed perfect truss using method of joints and Method of section.	**5 Hours**	**11%**
5	**Centroid and Centre of Gravity** Introduction, basic definitions and their understanding, Concept of centre of gravity, Centroids of Linear elements & Planar elements, Determination of centroids by integrations, Centroids of Composite sections (1D & 2D).	**6 Hours**	**14%**
6	**Fundamentals of Kinematics and Kinetics of Particles** Rectilinear motion, Curvilinear motion, Motion of rigid bodies, Velocity and acceleration, Newton's law of motion, Energy and momentum.	**5 Hours**	**11%**

Course Outcomes (COs):

On the completion of the course one should be able to:

CO1	Understand laws of mechanics and their application to engineering problems.
CO2	Understand the fundamentals of statics and apply them to simple structural problems
CO3	Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple practical problems.
CO4	Compute centroid of composite sections.
CO5	Recognize different types of friction and its effects on an object.
CO6	Classify different types of truss and analyze the pin jointed perfect truss.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO 5	PO 6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PS O2	PSO 3
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	1	-	-	-	-	-	-	-	1	-	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	1	-	-
CO4	2	1	1	-	-	-	-	-	-	-	-	-	1	-	-
CO5	1	1	-	-	-	-	-	-	-	-	-	-	1	-	-
CO6	1	1	2	-	1	-	-	-	-	-	-	1	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

❖ **Text Books:**

1. Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I & II, Charotar Publishing House
2. Junnarkar, S. B. & Shah, H. J., Applied Mechanics, Charotar Publishing House
3. Beer and Johnston, Engineering Mechanics (Statics & Dynamics)

- **Reference Books:**

1. Beer and Johnston, Mechanics of Materials
2. Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
3. Hibbler, R.C., Engineering Mechanics, Pearson Education
4. Popov, E.P., Engineering Mechanics of Solids, Prentice Hall of India, New Delhi
5. Meriam, J. L. & Kraige, L. G., Engineering Mechanics Statics, John Wiley & Son,
   Singapore
6. A K Tayal, Engineering Mechanics (Statics & Dynamics), Umesh Publications

- **Web Materials:**

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm
2. http://nptel.iitm.ac.in/video.php?subjectId=105106116

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICS
PY141.01: ENGINEERING PHYSICS

Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	**4**
Marks	100	50	-	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Error Analysis	05
2	Introduction to Quantum Mechanics	08
3	Acoustics and Ultrasonic	08
4	Physics of Laser and its applications	09
5	Condensed Matter and Material Physics	10
6	Nanoscience	05

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

B. Detailed Syllabus:

1. Error Analysis: 05 hours 11%

1.1 Uncertainties in Measurements: Sources and estimation of errors, accuracy and precision, systematic error, random error, Significant figure and round off

1.2 Uncertainties, Parent and Sample Distributions, Mean and Standard Deviation of Distributions

1.3 Average error, r.m.s error, probable error and error propagation

1.4 Introduction to regression analysis and its uses

1.5 Applications with numericals

2. Introduction to Quantum Mechanics: 08 hours 18%

2.1 Origin of quantum mechanics

2.2 Dual nature of matter, concept of wave group, Davisson and Germer experiment, 2.3 Heisenberg's uncertainty principle and its applications, Schrodinger's wave equation

2.4 Interpretation of wave equation and wave function

2.5 Applications of Schrodinger's wave equation

3. Acoustics and Ultrasonic: **08 hours 18%**

3.1 Introduction to waves

3.2 Characteristics of Sound waves

3.3 Architectural Acoustics, Absorption Coefficient, Reverberation, Sabine's formula, Factors affecting acoustics of buildings and their remedies

3.4 Ultrasonic: Introduction and properties

3.5 Production: piezoelectric and magnetostriction method

3.6 Detection of Ultrasonic waves

3.7 Applications with numericals

4. Physics of Laser and its applications: **09 hours 20%**

4.1 Lasers and its properties, Spontaneous and stimulated emission

4.2 Einstein coefficients

4.3 Gas laser (He-Ne Laser), Semiconductor Laser, Applications of Laser

4.4 Basic Principle of Holography, Construction and reconstruction of hologram, Applications of Holography

4.5 Introduction to optical fibre, Numerical Aperture of optical fibre

4.6 Types of optical fibre, applications of optical fibre

5. Condensed Matter and Material Physics: **10 hours 22%**

5.1 Basics of Crystal structure

5.2 X – Ray: properties, production, applications of X – Rays

5.3 Conductors, Insulators and Semiconductors: Band theory of Solids

5.4 Energy gap, Fermi energy, Electrical conductivity and mobility

5.5 Hall effect

5.6 Introduction to magnetic materials

5.7 Basics of superconductivity and its applications

6. Nanoscience: **05 hours 11%**

6.1 Introduction to nanomaterials

6.2 0D, 1D, 2D nanostructures

6.3 Bottom-up and Top-down approach to synthesis of nanomaterials

6.4 Structural characterization by XRD and SEM

6.5 Applications of nanomaterials

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 10 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.

- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	The student would be able to apply the concepts of physics in various branches of engineering.
CO2	An ability to identify, formulate and solve engineering problems.
CO3	An ability to use the techniques, skills and modern tools of physics necessary for engineering applications.
CO4	An ability to conduct experiments, analyse and interpret data.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2	-	-	-	-	-	-	-	-
CO2	3	3	3	1	2	-	-	-	-	-	-	-
CO3	3	3	-	3	3	-	-	-	-	-	-	-
CO4	3	3	2	3	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)
If there is no correlation, put "-"

Recommended Study Material:

- **Text Books:**
  1. Vijayakumari, G., Engg. Physics, Vikas Publishing house Pvt. Ltd.
  2. Rajagopal, K., Engg. Physics, Prentice Hall of India Pvt. Ltd.
  3. Avadhalula, M. N. & Kshirsagar, P. G., A text book of Engg. Physics, S. Chand Pub.
  4. Engineering Physics by D. R. Joshi
- **Reference Books:**
  1. Data Reduction and Error Analysis for the Physical Sciences 3rd Ed by P. R. Bevington & D Keith Robinson, McGraw – Hill (2003)
  2. Errors of Observation and their Treatment by J Topping
  3. An introduction to Error Analysis 2nd Edition by John R.Taylor, University Science Books
  4. Numerical methods for scientists and engineers by H M Antia, Tata McGraw-Hill
  5. Concept of Modern Physics by Beiser Arthur
  6. Introduction to Quantum Mechanics by D. J. Griffiths
  7. A text book of Quantum Mechanics, by P.M. Mathews and K. Venkatesan (TMH)
  8. Waves and Oscillations by Crawford F.S., Berkeley Physics Course, McGraw-Hill
  9. Nayak Abhijit, Engg. Physics, S. K. Kataria and Sons Pub.
  10. Kittle, C., Solid State Physics
  11. Fundamental of Physics by Haliday, Resnick and Walker
  12. Optics by Ghatak, Tata McGraw Hill, 3rd Edition
  13. Optics by E. Hecht
  14. Pillai, S.O., Solid State Physics, Wiley Eastern Ltd.
  15. Nano: The Essentials by T. Pradeep
  16. Textbook of Nanoscience and Nanotechnology by B. S. Murty, Baldev Raj, James Murday, and P. Shankar

- **Web Materials:**
  1. http://www.nptel.iitm.ac.in/courses/Webcourse-contents/IIT %20Guwahati/engg_physics/index_cont.htm
  2. http://ncert.nic.in/html/learning_basket.htm
  3. http://physics-animations.com/Physics/English/optics.htm
  4. http://physics-animations.com/Physics/English/waves.htm
  5. http://www.epsrc.ac.uk
  6. http://www.pitt.edu/~poole/physics.html#light

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS101.01 A: PAINTING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**2**
Marks	-	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Painting	02
2	Drawing from Nature and Object	04
3	Colour Design and Colour Value	06
4	Composition and Perspective	06
5	Figure Drawing and Proportion	04
6	Sketching	04
7	Contemporary Issues in Painting	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

1. An Introduction to Painting **02 Hours 06%**

- *An Introduction to Painting*
- *Principles of Composition*
- *Medium and Techniques of Painting*
- *History of Painting: Folk Indian Painting / Western Painting*
- *2D and 3D Painting*

2. Drawing from Nature and Object **04 Hours 12%**

- *Objects of Drawing: Nature and Manmade / Artificial Objects*
- *Drawing Still / Live Objects*
- *Drawing from Memory*
- *Drawing from Life*

3. Light and Shade 06 Hours 18%

- ***Color Theory**:*
  Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)
- ***Color Contrast & Attributes**:*
  Interaction, harmony, psychology/mood, culture & expression
- ***Media Characteristics & Surfaces**:*
  Acrylic, oil, paper, wood & canvas (primed/unprimed)

4. Composition and Perspective 06 Hours 18%

- ***Composition:***
  Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting
- *Perspective: An approximate reproduction*
- ***Types of Perspectives:***
- *Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective*

5. Figure Drawing and Proportion 04 Hours 12%

- *Proportions of the Human Body*
- *Three views –* ***Anterior (front), Lateral (side) and Posterior (back)***
- *Fundamental Proportion – The Big Three*

6. Sketching 04 Hours 12%

- *Sketching and Freehand*
- *Sketching Techniques*
- *Sketch and Drawing Medium*

7. Contemporary Issues in Painting 04 Hours 12%

- *Contemporary Indian Art*
- *Pioneers of Contemporary Indian Art*
- *Contemporary Issues in Painting*

C. Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.

CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	2	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

❖ **Reference Books**

1. A.N. Hodge (2007) *The History of Art: The Essential Guide to Painting Through the Ages* Arcturus Foulsham, London
2. David Lewis (1984) Pencil *Drawing Techniques* Watson-Guptill Publications, New York
3. John C Van Dyke L H D (1895) *A Text Book of the History of Painting* Longmans Green and Co., London
4. Sarah Parks (2014) Drawing *Secrets Revealed - Basics How to Draw Anything-* North Light Books, Ohio

❖ **Web Material**

1. https://conceptartempire.com/what-are-the-fundamentals/

2. https://www.creativebloq.com/art/painting-techniques-artists-31619638
3. https://www.thesprucecrafts.com/perspective-in-paintings-2578098
4. https://www.thehansindia.com/posts/index/Hans/2016-05-19/Origin-of-painting-in-India/229138

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS102.01 A: PHOTOGRAPHY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**2**
Marks	-	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Photography	03
2	Camera and Operating System	05
3	Light and Shade	10
4	Composition	09
5	Contemporary Issues in Photography	03

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

1. An Introduction to Photography **03 Hours 10%**

- *Art, Design and Visualization*
- *Basics of Photography and Various Types of Photography*
- *Basics of Post Production*
- *A Brief History of Photography:*
- *Early Experiments and Later Developments*

2. Camera and Operating System **05 Hours 16%**

- *Role of Camera in the Photography*
- *Types of Camera*

Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera

- *Principal Parts of Photographic Camera*
  Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer
- *Types of Lenses*
  Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses
- *Different Models of Camera, their Features and Operating Systems*
- *Camera and Size of the Image, Speed and Power of Lens*

3. Light and Shade **10 Hours 34%**

- *Reflection and refraction of light*
- *Dispersion of light through a glass prism, lenses*
- *Colour Filters:*
  Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter
- *Photographic Light Sources:*
  Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions
- *Artificial light sources:*
  Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands
  Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table

4. Composition **09 Hours 30%**

- *Different kinds of image formations*
- *Principal focus and focal length of the lens*
- *Depth of field, angle of view and perspective*
- *Perspective and composition*
- *Rules of composition*

5. Contemporary Issues in Photography **03 Hours 10%**

- *Present Day Photography*
- *Contemporary Photographers and their Contributions*
- *Major Issues in Contemporary Photography*

C. Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography

CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	2	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	2	-	-

Recommended Study Material:

❖ **Reference Books**

1. Maria Short, Sri-Kartini Leet and Elisavet Kalpaxi (2020) *Context and Narrative in Photography*-Routledge, New York.
2. Kevin Los (2010) *Improve Your Photography: 50 Essential Digital Photography Tips & Techniques* Digital Photography Series, New York.

❖ **Web Material**

1. https://photographylife.com/photography-basics
2. https://artofvisuals.com/the-basics-of-photography-introduction-to-photography-tutorials/
3. https://www.photographytalk.com/beginner-photography-tips/camera-basics-for-beginners
4. https://www.makeuseof.com/tag/5-photography-tips-for-beginners/
5. https://annamcnaught.com/blog/basics-of-photography

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS105.01 A: MEDIA AND GRAPHIC DESIGN

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**2**
Marks	-	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Media and Graphic Design	03
2	Layout and Design	07
3	Form and Space	06
4	Computer Graphics	04
5	Fonts	04
6	Basic Print Media	03
7	Contemporary Issues in Graphic Design	03

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

1. An Introduction to Media and Graphic Design — 03 Hours 10%

- *Creating Art, Art in Context and Art as Inquiry*
- *History of Graphic Design*
- *Constructional, Representational, and Simplification Drawing*

2. Layout and Design — 07 Hours 23%

- *Layout, Design and Aesthetics*
- *Elements of Design*
- *Principles of Design:*

Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion

- *Impact/function of Design*
- *Indigenous design practices*
- *Role of design in the changing social scenario*

3. Form and Space **06 Hours 20%**

- *Types of Forms: Man-made, Nature*
- *Types of Space: Negative and Positive*
- *Composition of Form and Space to create Layout Exploring Creativity*

4. Computer Graphics **04 Hours 12%**

- *An Introduction to Graphic Software*
- *Flash, Coreldraw, Illustrator and Photoshop*
- *Pre-press Process*

5. Fonts **04 Hours 12%**

- *Construction of Type*
- *Anatomy of Type*
- *Visual Language*
- *Creating Logo and Symbol*

6. Basic Print Media **03 Hours 10%**

- *An Introduction to Press and its Development Phases*
- *Types of Press*
- *Types of Printing Technologies*
- *Post-press Processes*

7. Contemporary Issues in Graphic Design **03 Hours 10%**

- *Present Day Graphic Designs*
- *Contemporary Designers and their Contribution*
- *Major Contemporary Issues in Graphic Design*

C. Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.
CO6	have cultivated a sense of creativity.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-

CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	3	-	-	-	-	-	-	2	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Recommended Study Material:

❖ **Reference Books**

1. Ellen Lupton and Jennifer Cole Philips (2008) Graphic *Design: The New Basics*, Princeton Architectural Press, New York

2. Ryan Hembree (2011) *The Complete Graphic Designer: A Guide to Understanding Graphics and Visual Communication* Rockport Publishers, Beverly

❖ **Web Material**

1. https://99designs.com/blog/tips/graphic-design-basics/

2. https://edu.gcfglobal.org/en/beginning-graphic-design/fundamentals-of-design/1/

3. https://ultragraphicsmt.com/10-basic-concepts-to-improve-your-graphic-design/

4. https://www.jcsocialmedia.com/graphic-design/

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS109.01 A DRAMATICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**2**
Marks	-	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drama	06
2	History of Drama and Contemporary Theatre	06
3	Theatre Design and Techniques	06
4	Technicalities of Stage Performance	08
5	Contemporary Trends in Drama	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

1. Introduction to Drama **06 Hours 20%**

- Introduction to performing arts
- Drama - An art, a socializing activity, & a way of learning
- Form of Drama
- Elements of Drama
- Types of Drama

2. History of Drama and Contemporary Theatre **06 Hours 20%**

- Important world dramatists & drama–from Greek to modern
- Evolution of contemporary theatre in the context of developments in Indian theatre

- Major Movements in Drama

3. Theatre Design and Techniques **06 Hours 20%**

- Theatre Architecture
- Stage craft: Set, light, costume, make up, sound, props
- Theatre techniques: from selection of script to final performance

4. Technicalities of Stage Performance **08 Hours 26%**

- Selection of plot and character
- Improvisation
- Movement
- Voice, Speech, Imagination
- Character Development
- Scene Enactment

5. Contemporary Trends in Drama **04 Hours 14%**

- New Tendencies in theatre
- Drama and Society
- Using drama for Social Change and Education

C. Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	be aware about the concept of performing art and its nuances.
CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Recommended Study Material:

❖ **Reference Books**

1. Bridget Panet (2009) Essential Acting: A Practical Handbook for Actors, Teachers and Directors Routledge, New York.
2. Tom Bancroft (2012) Character Mentor: Learn by Example to Use Expressions, Poses, and Staging to Bring Your Characters to Life Focal Press, New York

❖ **Web Material**

1. https://entertainism.com/elements-of-drama
2. https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/5069/the%20elements%20of%20drama.pdf
3. https://marrzipandrama.co.nz/the-6-elements-of-drama/
4. https://www.backstage.com/magazine/article/theater-trends-broadway-2019-66726/

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS110.01 A || CONTEMPORARY DANCE

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	**2**
Marks	-	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to dance	04
2	Types of Dance	06
3	Basic Elements of Dance	04
4	Technical Skills in Professional Contemporary Dance	06
5	Contemporary Trends in Dance	10

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

1. Introduction to dance **04 Hours 14%**

- Dance as a Performing Art
- Dance as a Medium of Expression
- History and Development of Dance

2. Types of Dance **06 Hours 20%**

- Western dance and classical dance
- Salsa, rumba, hip hop, tap dance, belly dance, etc.
- Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc.
- Other Regional dance forms in India

3. Basic Elements of Dance **04 Hours 14%**

- Movements of different parts of a body for Expression
- Concepts of: Nritya, Laya and Taal

4. Technical Skills in Professional Contemporary Dance **06 Hours 20%**

- Dance technique: alignment, balance, co-ordination, flexibility and control
- Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc.
- Skills and processes of rehearsal and production:
- physical energy, stamina and athleticism

5. Contemporary Trends in Dance **10 Hours 32%**

- Prevalent trends and techniques in contemporary dance
- Future trends in contemporary dance form
- On Stage Performance

C. Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.
CO6	be able to develop ability to express through the form of dance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	2	-	-	-	-

Recommended Study Material:

❖ **Reference Books**

1. Helene Scheff, Marty Sprague and Susan Mc Greeve. (1939) *Experiencing dance from student to dance artist*. Human kinetics, New York

2. Robin Grove, Catherine J. Stevens, Shirley McKenzie (2005) Thinking in Four Dimensions: Creativity and Cognition in Contemporary Dance. Melbourne university press, Melbourne

❖ **Web Material**

1. https://www.liveabout.com/what-is-contemporary-dance-1007423#:~:text=Contemporary%20dance%20is%20a%20style,body%20through%20fluid%20dance%20movements.
2. https://dancemagazine.com.au/2014/01/whats-contemporary-dance-days/
3. https://www.frontiersin.org/articles/10.3389/fpsyg.2019.00071/full
4. https://www.thehindu.com/features/friday-review/dance/what-is-contemporary-dance/article5096022.ece

B. Tech. (ME/CL/EE) Programme

SYLLABI

(Semester – 2)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA144: Engineering Mathematics- II

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Total	Credit
Hours/week	4	1	5	4
Marks	100	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	First order and First degree Ordinary Differential Equations	08
2	Higher Order Ordinary Linear Differential Equations	12
3	Partial Differential Equations and Applications	10
4	Matrix Algebra –II	10
5	Improper and Multiple Integrals	10
6	Probability and Statistics	10

Total hours (Theory): 30 Hrs.

Practical Hours: 30 Hrs.

Total hours: 60 Hrs.

B. Detailed Syllabus:

1. First order and First degree Ordinary Differential Equations: 08 Hours 13%

1.1 Modeling of real world problems in terms of first order ODE
1.2 Concept of general and particular solutions
1.3 Initial value problems
1.4 Existence and Uniqueness of solutions by illustrations
1.5 Solutions of first order and first degree differential equations
1.6 Linear, Bernoulli, Exact and non-exact differential equations

2. Higher Order Ordinary Linear Differential Equations: 12 Hours 20%

2.1 Model of real world problems of higher order LDE
2.2 General Solution of Higher Order Ordinary Linear Differential Equations with Constant coefficients

2.3 Methods for finding particular integrals viz. variation of parameters and undetermined coefficients

2.4 LDE of higher order with variable coefficients: Legendre's Equations (Special case: Cauchy-Euler equation)

2.5 System of simultaneous first order linear differential equations

3.	**Partial Differential Equations and Applications:**	**10 Hours**	**17%**
3.1	Boundary valued problems		
3.2	Methods of solutions of first order PDE		
3.3	Lagrange's Linear Partial Differential Equations.		
3.4	Special types of Nonlinear PDE of the first order		
3.5	Solutions of Heat, Wave and Laplace equations using separation of variables.		
3.6	Modeling of real world problem in terms of PDE		
4.	**Matrix Algebra –II:**	**10 Hours**	**17%**
4.1	Revision of matrices, determinant		
4.2	Eigenvalues and Eigenvectors of matrices		
4.3	Eigenvalues and Eigenvectors of special matrices		
4.4	Cayley-Hamilton's Theorem and its applications.		
4.5	LU decomposition		
5.	**Improper and Multiple Integrals:**	**10 Hours**	**17%**
5.1	Improper integrals and their convergence		
5.2	Definitions, properties and examples of Gamma, Beta and Error functions		
5.3	Evaluation of double and triple integrals		
5.4	Change of order of double integration		
5.5	Transformation to polar and cylindrical coordinates		
5.6	Applications of double and triple integrals		
6.	**Probability and Statistics:**	**10 Hours**	**16%**
6.1	Mean, median, mode and standard deviation		
6.2	Combinatorial probability		
6.3	Joint and Conditional probability and Bayes theorem		
6.4	Random variables, probability distribution functions - Binomial, Poisson, exponential and normal.		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Quiz (surprise test) /Oral tests/ Viva/Assignment/Tutorials will be conducted which carries 10% component of the overall evaluation.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	formulate models of natural phenomena using differential equations and find its solution using standard methods.
CO2	identify, analyze and subsequently solve physical problems analytically whose behaviour can be described by linear and nonlinear differential equations.
CO3	find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix, construct LU decomposition of a square matrix.
CO4	use advanced techniques to evaluate improper integrals, apply multiple integrals to find area, volume and mass in engineering field.
CO5	recognize the difference between different measure of central tendency, summarize and interpret data.
CO6	understand and solve the problems using probability axioms, rules and Bayes theorem, use distributions such as Binomial, Poisson, Exponential and Normal to solve real world problems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	2	-	1	1	-	-	-	-	-	-	-	-	-
CO6	2	2	1	2	1	1	1	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- ❖ Text Books:
  1. Erwin Kreyszig; Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
  2. H. K. Dass and Rajnish Verma; Higher Engineering Mathematics, S Chand & Co Pvt. Ltd.
  3. Sheldon Ross; A first course in probability. Pearson, 2014.
  4. B. S. Grewal; Higher Engineering Mathematics, Khanna Publ., Delhi, 2012

❖ Reference Books:

1. M. D. Weir et al*;* Thomas' Calculus, 11^{th} Ed., Pearson Education, 2008.
2. James Stewart; Calculus Early Transcendental, 5^{th} Ed., Thomson India, 2007
3. C. R. Wylie and L. C. Barrett; Advanced Engineering Mathematics. 1982, McGraw-Hill Book Company.
4. Michael D. Greenberg; Advanced engineering mathematics. Prentice-Hall, 1988.
5. R. V. Hogg, E. A. Tanis and D. L. Zimmerman; Probability and Statistical Inference, 9th edition, Prentice Hall, 2015.
6. Zafar Ahsan; Differential Equations and Their Applications, φ Learning, Pvt Ltd,Third Edition (2017).

❖ URL Links:

1. http://nptel.ac.in/courses/122107037/
2. http://nptel.ac.in/courses/111107108/
3. http://nptel.ac.in/courses/122103012/
4. http://nptel.ac.in/courses/122104018/
5. http://nptel.ac.in/courses/111106100/
6. http://nptel.ac.in/courses/122101003/
7. https://ocw.mit.edu/courses/mathematics/18-02-multivariable-calculus-fall-2007/lecture-notes/
8. https://nptel.ac.in/courses/111105041/

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL ENGINEERING
ME 147: BASICS OF CIVIL & MECHANICAL ENGINEERING
(Introduced in 2019-20)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	**4**
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction of Mechanical Engineering	06
2.	Steam and Steam Generator	05
3.	Internal Combustion Engines	06
4.	Pumps and Compressors	03
5.	Refrigeration and Air Conditioning Systems	05
6.	Power and Motion Transmission	05
7.	Scope of Civil Engineering	02
8.	Introduction to Surveying	06
9.	Elements of building Construction	07

Total hours (Theory): 45 Hrs.

Practical Hours: 30 Hrs.

Total hours: 75 Hrs.

B. Detailed Syllabus:

PART A:

1. Introduction of Mechanical Engineering **06 Hours** **13%**

1.1 Prime movers and its types, sources of energy

1.2 Basic terminology: force and mass, pressure, work, power, energy, heat, temperature, units of heat, specific heat capacity, interchange of heat, change of state, internal energy, enthalpy, entropy, efficiency

1.3 Zeroth law and first law of thermodynamics, Boyle's law, Charle's law and combined gas law, Relation between Cp and Cv

1.4 Constant volume process, constant pressure process, isothermal process, poly-tropic process, adiabatic process

1.5 Numerical practice

2.	**Steam and Steam Generator**	**05 Hours**	**11%**
2.1	Introduction to steam formation and its types		
2.2	Introduction to steam table		
2.3	Calorimeter and its types		
2.4	Boiler definition and its classification		
2.5	Cochran boiler, Babcock and Wilcox boiler and its mountings and accessories		
2.6	Efficiencies of boiler		
3.	**Internal Combustion Engines**	**06 Hours**	**13%**
3.1	Introduction		
3.2	Basic terminology of I.C. engine		
3.3	Types of I. C. engines- four stroke & two stroke engines		
3.4	Efficiencies of an engine		
3.5	Numerical practice		
4.	**Pumps and Compressors**	**03 Hours**	**7%**
4.1	Introduction		
4.2	Classification and application of pumps and compressors		
4.3	Working of reciprocating and centrifugal type pumps & compressors		
5.	**Refrigeration and Air Conditioning Systems**	**05 Hours**	**11%**
5.1	Introduction to refrigeration and air conditioning		
5.2	Basic terminology, principal and application of refrigeration		
5.3	Vapour compression refrigeration system		
5.4	Domestic refrigerator		
5.5	Window and split air conditioning systems		
6.	**Power and Motion Transmission**	**05 Hours**	**11%**
6.1	Introduction		
6.2	Types of couplings, brakes and clutches		

6.3 Belt drive and its types

6.4 Gear drives and its types, gear trains, chain drives

PART B:

7. Scope of Civil Engineering — 02 Hours — 05%

7.1 Scope of Civil engineering

7.2 Branches of civil engineering,

7.3 Role of civil engineer

8. Introduction to Surveying — 06 Hours — 13%

8.1 Definition of surveying

8.2 Objects of surveying, uses of surveying

8.3 Primary divisions of surveying, principles of surveying

8.4 List of classification of surveying, definition: plan and map

8.5 Introduction to linear and angular measurements

9. Elements of building Construction — 07 Hours — 16%

9.1 Types of building

9.2 Design loads

9.3 Building components (super structure and substructure)

9.4 Principles of planning

9.5 Basics requirements of a building planning

9.5 Types of residential building

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

- In the lectures and laboratory discipline and behaviour will be observed strictly.

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand fundamental principles of mechanical engineering
CO2	Learn the formation of steam and different types of the boilers and its mounting and accessories
CO3	Explore the basics of internal combustion engine and pumps & compressors
CO4	Learn basics of refrigeration and air conditioning system and power & motion transmission systems
CO5	Learn importance and application of civil engineering
CO6	Understand the fundamentals of surveying and building planning and Identify different building components

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	1	-	-	-	-	-	-	-
CO5	2	-	-	-	-	1	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	1	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- ❖ **Text book:**
    1. S.M.Bhatt, H.G.Katariya, J.P.Hadiya, “Elements of Mechanical Engineering”, Books India Publication, Ahmedabad.
    2. P.S.Desai, S.B.Soni, “Elements of Mechanical Engineering”, Atul Prakashan, Ahmedabad
    3. Kandya Anurag, “Elements of Civil Engineering”, Charotar Publishing House Pvt. Ltd.
- ❖ **Reference book:**
    1. Dr. Sadhu Singh, “Elements of Mechanical Engineering”, S.CHAND Publication, New Delhi
    2. V.K.Manglik, “Elements of Mechanical Engineering”, PHI Learning, Delhi.
    3. Khasia R.B. and Shukla R.N., “Elements of Civil Engineering”, Mahajan Publication.
    4. Punamia B.C., “Surveying”, Vol. I & II
- ❖ **Web Materials:**
    1. http://nptel.ac.in/downloads/112105125/
    2. http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101
    3. http://nptel.ac.in/courses/105107122/
    4. https://swayam.gov.in/nd1_noc19_me58/preview

FACULTY OF TECHNOLOGY AND ENGINEERING
M. S. PATEL DEPARTMENT CIVIL ENGINEERING
CL142.01: ENVIRONMENTAL SCIENCE

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	05
2	Environmental Pollution	12
3	Ecology &Ecosystems	10
4	Natural Resources	03

Total Hours (Theory):00 Hrs.

Total Hours (Lab):30 Hrs.

Total Hours: 30 Hrs.

B. Detailed Syllabus:

1 Introduction 05 Hours 24%

1.1 Basic definitions

1.2 Objectives and guiding principles of environmental studies

1.3 Components of environment

1.4 Structures of atmosphere

1.5 Man-Environment relationship

1.6 Impact of technology on the environment

2 Environmental Pollution 12 Hours 33%

2.1 Environmental degradation

2.2 Pollution, sources of pollution, types of environmental pollution

2.3 Air pollution: Definition, sources of air pollution, pollutants, classification of air pollutants (common like SO_X &NO_X), sources &effects of common air pollutants

2.4 Water pollution: Definition, sources water pollution, pollutants &classification of water pollutants, effects of water pollution, eutrophication

2.5 Noise pollution: Sources of noise pollution, effects of noise pollution

2.6 Ill Effects of Fireworks: Severity of toxicity, environmental effects and health hazards.

2.7 Current environmental global issues, global warming &green houses, effects, acid rain, depletion of Ozone layer

3 Ecology &Ecosystems **10 Hours 33%**

3.1 Ecology: Objectives and classification

3.2 Concept of an ecosystem: Structure &function

3.3 Components of ecosystem: Producers, consumers, decomposers

3.4 Bio-Geo-Chemical cycles &its environmental significance

3.5 Energy flow in ecosystem

3.6 Food Chains: Types &food webs

3.7 Ecological pyramids

3.8 Major ecosystems

4 Natural Resources **03 Hours 10%**

4.1 Natural resources: Renewable resources, nonrenewable resources, destruction versus conservation

4.2 Energy resources: Conventional energy sources& its problems, non-conventional energy sources-advantages &its limitations, Problems due to overexploitation of energy resources

Course Outcomes (COs):

At the end of the course, the students will be able:

CO1	To perceive the elementary knowledge about natural environment and its relation with science.

CO2	To identify and analyze human impacts on the environment.
CO3	To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.
CO4	To Initiate new and innovative environmental friendly practices.
CO5	To communicate on recent environmental problems thereby creating awareness among society.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO3	-	-	-	-	-	2	3	-	-	-	-	-	-	3
CO4	-	-	2	-	-	-	3	-	-	-	-	-	2	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text Books:**

1. Varandani,N.S., Basics of Environmental Studies
2. Sharma, J.P., Basics of Environmental Studies

❖ **Reference Books:**

1. Shah Shefali& Goyal Rupali, BasicsofEnvironmentalStudies
2. Agrawal,K.C., Environmental Pollution: Causes, Effects &Control
3. Dameja,S.K., Environmental Engineering& Management
4. Rajagopalan,R., Environmental Studies, Oxford University Press
5. Wright Richard T.& Nebel Bernard J., Environmental Science
6. Botkin Daniel B.& Edward A. Keller, Environmental Science
7. Shah,S.G.,Shah,S.G.&Shah,G.N.,BasicsofEnvironmentalStudies,SuperiorPublications,Vadodara

- **Web Materials:**

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Environmental%20Air%20Pollution/index.htm
2. http://nptel.iitm.ac.in/video.php?subjectId=105104099
3. http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html
4. http://www.epa.gov
5. http://www.globalwarming.org.in
6. http://nopr.niscair.res.in
7. http://www.indiaenvironmentportal.org.in

FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE145: Basics of Electronics & Electrical Engineering

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	**4**
Marks	100	50	0	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Electrical Terms and Units	04
2	Electrical Circuit Analysis	07
3	Electrostatic	08
4	Electromagnetism	05
5	AC Fundamentals	05
6	Single Phase AC Series Circuits	05
7	Polyphase Circuits	04
8	Basics of Electronics	07

Total hours (Theory): 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

B. Detailed Syllabus

1. Basic Electrical Terms and Units **04 Hours** **08%**

1.1 Ohm's law, resistor and its coding, properties, temperature co- efficient of resistance, resistance variation with temperature, examples

2. Electrical Circuit Analysis **07 Hours** **15%**

2.1 Kirchhoff's current and voltage law, mesh and nodal analysis, Examples

2.2 Series parallel circuits, star-delta transformation

3. Electrostatic **08 Hours** **18%**

3.1 Capacitors, charge and voltage, capacitance, electric fields, electric field strength and electric flux density, relative permittivity, dielectric strength, Examples

3.2 Capacitors in parallel and series, Calculation of capacitance of parallel plate and multi plate capacitor, examples

4.	**Electromagnetism**	**05 Hours**	**12%**
4.1	Magnetic field, its direction and characteristics, magnetic flux and flux density, magneto motive force and magnetic field strength, examples		
4.2	Faraday's law of electromagnetic induction, Fleming's left hand and right hand rule, Lenz law, force on a current carrying conductor, examples		
4.3	Self and mutual inductance		
5.	**AC Fundamentals**	**05 Hours**	**12%**
5.1	AC Waveform and definition of its terms, relation between speed and frequency		
5.2	Average and RMS value and its determination for sinusoidal wave shapes, examples		
6.	**Single Phase AC Series Circuits**	**05 Hours**	**12%**
6.1	R–L and R-C series circuit, power in ac circuits, examples		
6.2	R-L-C series circuit, resonance in R-L-C series circuit, relevant examples		
7.	**Polyphase Circuits**	**04 Hours**	**08%**
7.1	Phase sequence, voltage and current relations in star and delta connected system		
8.	**Basics of Electronics**	**07 Hours**	**15%**
8.1	Electronic Systems: Basic amplifier, voltage, current and power gain, Basic attenuators, CRO		
8.2	Transmission and Signals: Analog and digital signals, bandwidth,		
8.3	Forward and reverse bias of PN junction diode, Zener diode		
8.4	Rectifiers: Half Wave, Full Wave - Centre Tap, Bridge		
8.4	Transistor: Bipolar junction transistor, construction and biasing, configuration		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, blackboard, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

Course Outcomes (COs):

On the completion of the course, one should be able to:

CO1	Describe resistors, capacitors and inductors properties, readings and calculation.
CO2	State the basic electrical laws and apply these laws to solve electrical network.
CO3	Identify the property of magnetic materials and understand the laws of emf generation.
CO4	Solve the series and parallel DC circuits and AC circuits for single and poly-phase networks.
CO5	Develop skill and design AC-DC rectification circuits, operate basic electrical and electronics instruments.

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	1	-
CO3	1	3	-	-	-	-	-	-	-	-	-	-	1	-
CO4	1	2	-	2	-	-	-	-	-	-	-	-	3	-
CO5	1	2	3	-	3	-	-	-	-	-	-	-	3	-

Recommended Study Material:

❖ **Text Books:**

1. Elements of Electrical Engineering and Electronics by U.A.Patel and R. P. Ajwalia
2. A Text Book of Electrical Technology by B. L. Thareja, S. Chand
3. Principles of Electrical Engineering and Electronics by V. K. Mehta, S. Chand

❖ **Reference Books:**

1. Electrical Technology by Hughes, Pearson Education.
2. Electrical Engineering Fundamentals by Vincent Del Toro, Pearson Education.

❖ **Web Material**

1. https://www.electronics-tutorials.ws/

FACULTY OF TECHNOLOGY & ENGINEERING
Smt. Kundanben Dinsha Patel Department of Information Technology
IT143: Fundamentals of Computer Programming

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	0	4	**3**
Marks	100	50	0	150	

A. Outline of the course

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Computation and Algorithm	03
2	Introduction to Programming and 'C' language	02
3	Constants, Variables & Data Types in 'C'	03
4	Operators and Expression in 'C'	03
5	Managing Input & Output Operations	02
6	Conditional Statements, Branching & Looping	04
7	Arrays	04
8	Character Arrays	03
9	User-Defined Functions in 'C'	03
10	Structures and Basics of pointer	03

Total hours (Theory): 30 Hrs.

Total hours (Lab): 30 Hrs.

Total hours: 60 Hrs.

B. Detailed Syllabus:

The following contents will be delivered to the students during laboratory sessions.

1 **Introduction to Computation and Algorithm** **03 Hours 9%**

Various number systems: Decimal, Binary, Octal, Hexadecimal, conversion from one number system to another, What is computer, Algorithms, Flow-charts.

Solve Various types of algorithms like Exchanging values of two variables, (using 3 variables & 2 variables), Arrange numbers in ascending order, Evaluate various series e.g.: sin x, $1^2-2^2+3^2-\ldots$, $1+2/2!+3/3!+\ldots$,

2	**Introduction to Programming and 'C' language**	**02 Hours**	**7%**
	What is program & programming, programming languages and it's types, compiler, and interpreter, History of C, Characteristics of C, Basic structure of a program, Compiling process of C a Program		
3	**Constants, Variables & Data Types in 'C'**	**03 Hours**	**9%**
	Character set, C tokens, Keyword, Constants, and Variables Data types – declaration & initialization, User-defined type declaration - typedef, enum, Symbolic constant (#define)		
4	**Operators and Expression in 'C'**	**03 Hours**	**13%**
	Classification of operators: arithmetic, relational, logical, assignment, increment / decrement, bitwise, special operators. Unary, binary and ternary operators, Arithmetic expression, evaluation, type conversion: implicit & explicit, precedence and associativity, use of math.h file		
5	**Managing Input & Output Operations**	**02 Hours**	**6%**
	Basic input and output operations, Input a character, introduction to ASCII code, various library functions from ctype.h. Formatted input using scanf( ), formatted output of integer and real data using printf ( )		
6	**Conditional Statements, Branching & Looping**	**04 Hours**	**14%**
	Decision making using if, if…else statement, nesting of if…else, else…if Ladder. *switch*, use of if…else instead of conditional operator, *goto* statement Need of looping, entry-controlled loop: while, for, exit-controlled loop: do…while, difference, Nesting of looping statements, use of break and continue, use of if, if…else in loop		
7	**Arrays**	**04 Hours**	**10%**
	Need of array, declaration & initialization 1D array, various programs of 1D		

	2D array and their memory allocation, 2D array basic programs and matrix operations		
8	**Character Arrays**	**03 Hours**	**8%**
	Difference of character array with numeric array and importance of NULL character, Declaration, initialization and various input and output methods of string.		
9	**User-Defined Functions in ‘C’**	**03 Hours**	**12%**
	Need of modularization, advantages, introduction to user-defined function, form of C functions, function prototype, function call, function body, Call by value, actual & formal arguments, use of return, nesting of functions, recursion, Array as function arguments, storage classes: scope, life of a variable in C		
10	**Structures and Basics of pointer**	**03 Hour**	**12%**
	Need of user-defined data type, structure definition, declaration and initialization of variables. Background of memory, variable, value, address of variable, introduction to pointer, declaration & initialization, access value using pointer, indirection (*) operator		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed. Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory and tutorial session respectively.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify situations where computational methods would be apply and develop various problem solving skills.
CO2	Define, select, and compare data types, operators, and functions to solve mathematical as well as scientific problems.
CO3	Describe, design and develop modular programs using control structures.
CO4	Illustrate the use of different data structures with practical aspect.
CO5	Validate the logic building and code formulation by designing code capable of passing various test cases.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	1	1	1	-	-	-	-	-	-	-	-	-	1	-
CO3	2	-	-	-	1	-	-	-	-	-	-	-	1	-
CO4	-	-	-	1	-	-	-	-	-	-	-	-	1	1
CO5	2	1	2	-	-	-	-	-	-	-	-	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- ❖ **Text book:**
    1. Programming in ANSI C, E.Balagurusamy, Tata McGraw Hill.
- ❖ **Reference book:**
    1. C Programming Language (2nd Edition), Brian W. Kernighan, Dennis M. Ritchie, Prentice-Hall (PHI)
    2. C: The Complete Reference, Herbert Schildt, McGrawHill
    3. Let us C: Yashwant Kanetkar, BPB publications new delhi

4. Computer programming and utilization: M.T.Savaliya, Atul Prakashan
5. Computer concepts and Programming , Vikas Gupta, DreamTech
6. Computer fundamentals and Programming in C, Pradip dey and Manas Ghosh, Oxford.

❖ **Web Materials:**

1. http://www.tutorials4u.com/c/
2. http://www.cprogramming.com/tutorial.html
3. http://www.howstuffworks.com/c.htm
4. http://www.programmingtutorials.com/c.aspx
5. http://www.physics.drexel.edu/courses/Comp_Phys/General/C_basics/

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS126.01 A: Communication Skills 1

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	-	-	-	**2**
Marks	100	-	-	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Communicative English	04
2	Functional English	04
3	Conversational Skills	04
4	Introduction to Communication and Key Concepts in Communication	06
5	Effective Listening and Reading Skills	06
6	Writing Skills	06

Total hours (Theory): 30 Hrs.

B. Detailed Syllabus:

1. Communicative English **04 Hours 12%**
1.1 Time and Tenses
1.2 Active and Passive Voice
1.3 *Direct-Indirect Speeches*
1.4 Prepositions and Conditionals

2. Functional English **04 Hours 12%**
2.1 Introduction to Functional English
2.2 Describing Actions and Processes, Offering, Requests, Routines/Timetable, Making Comparisons, Sharing Interests and Experience

3. Conversational Skills **04 Hours 12%**

3.1 *An Introduction to Conversations for various purposes*

3.2 Importance of acquiring Conversational Skills

3.3 Models, Techniques and Types of Conversations

4. Introduction to Communication and Key Concepts in Communication **06 Hours 18%**

4.1 An Introduction to Communication

4.2 Basic Terms, Concepts, and Contexts of Communication

4.3 Factors influencing message encoding, the nature of message, and message uses and effects

4.4 Importance, Types and Principles of Communication

5. Effective Listening and Reading Skills **06 Hours 18%**

5.1 An Introduction to Listening and Reading

5.2 Purposes, Types and Techniques of Listening and Reading

5.3 Barriers to effective Listening & Reading and overcoming the Barriers

5.4 Note-taking and Note-making

6. Writing Skills **06 Hours 18%**

6.1 An Introduction to Writing

6.2 Importance of effective Writing

6.3 Paragraph Development: Coherence –Topic Sentence, Supporting Sentence & Data etc.

6.4 Business Letter Writing

C. Instructional Methods and Pedagogy:

- The course is based on pragmatic learning.
- Classroom Teaching will be facilitated by Reading Material, Classroom Discussions, Task-based learning, projects, assignments and
- Various interpersonal activities like case-studies, critical reading, group-work/pair-work, and presentations will be a part of the internal and External Assessment

Course Outcome (COs):

After completion of the course, the student would:

CO1	have polished their grammar and developed the ability to communicate effectively in business situations,
CO2	be able to communicate message accurately, handle situation that require thoughtful communication, to use appropriate words and tones and so on.
CO3	be able to use the language effectively for various functions
CO4	develop familiarity with varied styles of communication and gain insights into how to deal with people with different communication styles

CO5	hone basic linguistic and communication skills (of students) required in a business organization, namely: Listening, Speaking, Reading and Writing

Course Articulation Matrix:

At the end of the course, the student will be able to:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Enter Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no co-relation put "-"

Recommended Study Material

❖ **Reference Books:**

- Sanjay Kumar and Pushp Lata (First Edition, 2011), *Communication Skills*, Oxford University Press, New Delhi
- Krishna Mohan and Meena Banerji (2010), *Developing Communication Skills*, Macmillan Publications India Ltd., New Delhi
- M V Rodriques (2013), *Effective Business Communication*, Concept Publishing Company (P) Ltd., New Delhi
- Mohan and Meenakshi Raman (2006), *Effective English Communication Krishna,* Mcgraw - Hill Publishing Company Limited, New Delhi
- Geoffrey Leech & Jan Swartvik (1994), *A Communicative Grammar of English*, Longman Publications, New York
- Jones Leo (1979), *Functions of English*, Cambridge University Press, UK

❖ **Web material:**

- http://www.communicationskills.co.in/index.html
- http://www.hodu.com/default.htm
- http://www.bbc.co.uk/worldservice/learningenglish

- http://www.englishlearner.com/tests/test.html
- http://www.englishclub.com/vocabulary/idioms-body.htm
- http://dictionary.cambridge.org

Chandubhai S. Patel Institute of Technology
(Faculty of Technology & Engineering)

Division Allocation

Division	Group	Branch	Abbreviation	ID No.
1.	1	Civil Engineering	CL 1	19CL001 to 19CL060
2.			CL 2	19CL061 to 19CL120
3.		Mechanical Engineering	ME 1	19ME001 to 19ME060
4.			ME 2	19ME061 to 19ME120
5.		Electrical Engineering	EE 1	19EE001 to 19EE060
6.			EE 2	19EE061 to 19EE120
7.	2	Computer Engineering	CE 1	19CE001 to 19CE060
8.			CE 2	19CE061 to 19CE120
9.		Information Technology	IT 1	19IT001 to 19IT060
10.			IT 2	19IT061 to 19IT120
11.		Electronics & Communication	EC 1	19EC001 to 19EC060
12.			EC 2	19EC061 to 19EC120

Batch Allocation

Division	Batch	ID NO.	Division	Batch	ID NO.
1	A1	19CL001 to 19CL020	7	A1	19CE001 to 19CE020
	B1	19CL021 to 19CL040		B1	19CE021 to 19CE040
	C1	19CL041 to 19CL060		C1	19CE041 to 19CE060
2	A2	19CL061 to 19CL080	8	A2	19CE061 to 19CE080
	B2	19CL081 to 19CL100		B2	19CE081 to 19CE100
	C2	19CL101 to 19CL120		C2	19CE101 to 19CE120
3	A1	19ME001 to 19ME020	9	A1	19IT001 to 19IT020
	B1	19ME021 to 19ME040		B1	19IT021 to 19IT040
	C1	19ME041 to 19ME060		C1	19IT041 to 19IT060
4	A2	19ME061 to 19ME080	10	A2	19IT061 to 19IT080
	B2	19ME081 to 19ME100		B2	19IT081 to 19IT 100
	C2	19ME101 to 19ME120		C2	19IT101 to 19IT120
5	A1	19EE001 to 19EE020	11	A1	19EC001 to 19EC020
	B1	19EE021 to 19EE040		B1	19EC021 to 19EC040
	C1	19EE041 to 19EE060		C1	19EC041 to 19EC060
6	A2	19EE061 to 19EE080	12	A2	19EC061 to 19EC080
	B2	19EE081 to 19EE100		B2	19EC081 to 19EC100
	C2	19EE101 to 19EE120		C2	19EC101 to 19EC120

Faculty of Technology and Engineering
Chandubhai S Patel Institute of Technology
List of First Year Student Counselors

Student No.	Faculty Name	Seating Room No.	Intercom Number	Mobile Number
FY B TECH Civil Engineering				
A1 Batch: 19CL001 to 19CL020	Vipul Vyas	503	5085	9727767541
B1 Batch: 19CL021 to 19CL040	Pinal Patel	611	5090	9979975075
C1 Batch: 19CL041 to 19CL060	Dipali Patel	501	5084	9924999774
A2 Batch: 19CL061 to 19CL080	Dr. Hiteshri Shastri	627B	5092	8879137964
B2 Batch: 19CL081 to 19CL100	Jay Bhavsar	514	5088	9898583301
C2 Batch: 19CL101 to 19CL120	Neha Chauhan	512	5086	9099027974
FY B TECH Electrical Engineering				
A1 Batch: 19EE001 to 19EE020	Mihir Mehta	117B	5045	7405856957
B1 Batch: 19EE021 to 19EE040	Mihir Patel	124A	5047	9723315488
C1 Batch: 19EE041 to 19EE060	Jignesh Patel	124A	5047	9978794997
A2 Batch: 19EE061 to 19EE080	Mihir Bhatt	117B	5045	9913277673
B2 Batch: 19EE081 to 19EE100	Vibha Parmar	116	5044	9426501707
C2 Batch: 19EE101 to 19EE120	Kishan Patel	CSRTC	5135	9724442255
FY B TECH Mechanical Engineering				
A1 Batch: 19ME001 to 19ME020	Sagar Chokshi	523A	5329	9726622007
B1 Batch: 19ME021 to 19ME040	Madhav Oza	622	5234	8460236433
C1 Batch: 19ME041 to 19ME060	Kundan Patel	523A	5226	9624095919
A2 Batch: 19ME061 to 19ME080	Axat Patel	CSRTC	5135	9924991009
B2 Batch: 19ME081 to 19ME100	Gaurang Patel	620	5232	8141342100
C2 Batch: 19ME101 to 19ME120	Harmish Bhatt	616	5235	9428657212
FY B TECH Computer Engineering				
A1 Batch: 19CE001 to 19CE020	Mayuri Popat	702		9998976027
B1 Batch: 19CE021 to 19CE040	Martin Parmar	702		8488065457
C1 Batch: 19CE041 to 19CE060	Nikita Bhatt	623		8200937439
A2 Batch: 19CE061 to 19CE080	Bharat Sarkhedi	702		9978584838
B2 Batch: 19CE081 to 19CE100	Mrugendra Rahevar	702		9925856399
C2 Batch: 19CE101 to 19CE120	Ashwin Makwana	CDPC		9408151969
FY B TECH Information Technology				
A1 Batch: 19IT001 to 19IT020	Bimal Patel	601		9909428681
B1 Batch: 19IT021 to 19IT040	Chandani Shah	601		9824660719
C1 Batch: 19IT041 to 19IT060	Madhav Ajwalia	703		
A2 Batch: 19IT061 to 19IT080	Mrudang Pandya	601		9909828897
B2 Batch: 19IT081 to 19IT 100	Vishwa Vala	604		7600332117
C2 Batch: 19IT101 to 19IT120	Pritesh Prajapati	703		9558135477
FY B TECH Electronics & Communication				
A1 Batch: 19EC001 to 19EC020	Dr.Sarman K. Hadia	233-B	5073	9427062772
B1 Batch: 19EC021 to 19EC040	Brijesh N. Shah	224	5073	9426588167
C1 Batch: 19EC041 to 19EC060	Poonam J. Thanki	227	5064	9978992692
A2 Batch: 19EC061 to 19EC080	Jalpa H. Desai	227	5064	9898639553
B2 Batch: 19EC081 to 19EC100	Rajat G. Pandey	225	5071	9574715923
C2 Batch: 19EC101 to 19EC120	Yogesh S. Tiwari	205	5068	8347496808

Academic Calendar (2019) - ODD Semester (B. Tech: 1st Semester)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY, CHANGA									
FACULTY OF TECHNOLOGY AND ENGINEERING									
Week No	Month	M	T	W	T	F	S	S	Activity
1	July - August	15	16	17	18	19	H	21	Orientation Program Start (15/07/2019 to 27/07/2019)
2		22	23	24	25	26	27	28	Orientation Program (15/07/2019 to 27/07/2019) (cont..)
3		29	30	31	1	2	3	4	B Tech 1st Sem Academic Session starts (29/07/2019)
4	August	5	6	7	8	9	10	11	
5		DH	13	14	DH	16	H	18	
6		19	20	21	22	23	DH	25	
7		26	27	28	29	30	31	1	
8	Septmber	2	3	4	5	6	7	8	1st Notification of B Tech 1st Sem Attendance Report (07/09/2019)
9		9	10	11	12	13	14	15	B Tech 1st Sem 1st Sessional Exam (09/09/2019)
10		16	17	18	19	20	H	22	Result of B Tech 1st Sem 1st Sessional Exam (20/09/2019)
11		23	24	25	26	27	28	29	Workshop: Courses on Liberal Arts (Tentative: 25/09/2019 to 28/09/2019)
12	October	30	1	DH	3	4	5	6	Expert Lecture / Parents Teacher Meeting
13		7	DH	9	10	11	12	13	
14		14	15	16	17	18	H	20	
15		21	22	23	24	25	26	27	Diwali Break
16		28	29	30	31	1	2	3	Diwali Break
17	November	4	5	6	7	8	9	10	
18		11	12	13	14	15	H	17	2nd Notification of B Tech 1st Sem Attendance Report (15/11/2019)
19		18	19	20	21	22	23	24	B Tech 1st Sem 2nd Sessional Exam (18/11/2019)
20		25	26	27	28	29	30	1	Result of B Tech 1st Sem 2nd Sessional Exam (30/11/2019) B Tech 1st/2nd Sem Practical Exam (Tentative) (25/11/2019)
21	December	2	3	4	5	6	7	8	B Tech 1st/2nd Sem Practical Exam (Tentative) (cont..)
22		9	10	11	12	13	14	15	
23		16	17	18	19	20	H	22	B Tech 1st/2nd Sem (Regular/Backlog) Theory Exam (16/12/2019)
24		23	24	DH	26	27	28	29	B Tech 1st/2nd Sem (Regular/Backlog) Theory Exam (cont..)
25		30	31	1	2	3	4	5	B Tech 2nd Sem Start (30/12/2019)

[DH] - Declared Holidays (as per published list of holidays), [H] - Holiday

15/07/2019

Attendance Criteria and Leave Application

This is to inform all the students of **First Year B.Tech**.

- ☞ **Students should be regular and punctual to all the classes (Theory, Practical & Tutorials) and secure attendance of not less than 80% in each course. Student should be fully aware that attendance less than 80% in any of the course will make me ineligible to appear for the examination of that course.**
- ☞ Students should be fully aware that he/she shall not be allowed to enter the class if he/she is late.
- ☞ He/she will conduct himself/herself in a highly **disciplined** and decent manner both inside the classroom and in the campus, failing which suitable action may be taken against the student as per the rules and regulations of the College and University.
- ☞ Maximum duration of Leave during the semester is 15 Days only.
- ☞ Only Medical leaves in case of Major Surgery/Injury will be granted.
- ☞ In case of Major Surgery/Injury, immediate reporting to respective counselor is required.
- ☞ Students availing medical leaves need to submit Leave Report along with necessary documents to the **Attendance Coordinator of the respective class** within a week time. After that no leaves will be sanctioned.
- ☞ Students Leave Form is available at PA to Principal Office.
- ☞ Further, 100% Attendance is compulsory in Lab Sessions.

Dr A. D. Patel
Principal

Charotar University of Science and Technology
Faculty of Technology & Engineering [CSPIT/DEPSTAR]

*(Note: In order to make the students aware of the Academic re**gulations** and **attend** the classes regularly from the first day of starting of classes the following Undertaking Form is introduced which should be signed by both **student and parent**. The same should be submitted to the concerned HOD through Counsellor on the day of starting of semester classes)*

Undertaking by Students/Parents for Attendance/Academic Discipline

I, Mr/Ms____bearing ID No.__________ joining for I / II / IV / VI / VII/ VIII Semester, B.Tech/MTech (CL/ME/CE/CSE/IT/EC/EE) for the academic year ___________in **Faculty of Technology [CSPIT/DEPSTAR], Changa** do hereby undertake and abide by the following terms.

1. I will **attend** all the classes as per the time table from the **re-opening day** of the College on ______________I also understand that if I do not turn up even after two week of starting of classes, I shall be **ineligible** to continue for the current academic year.
2. **I will be regular and punctual to all the classes (Theory, Practical & Tutorials) and secure attendance of not less than 80% in each subject. I am fully aware that attendance less than 80% in any of the subjects will make me ineligible to appear for the examination of that subject and overall attendance less than 80% will make me ineligible to appear for the whole examinations.**
3. I am aware that the mere production of medical certificate will not exempt me from the minimum attendance criteria and all medical leaves are included in the overall 20% relaxation.
4. I am fully aware that I shall not be allowed to enter the class if I am late.
5. I will conduct myself in a highly **disciplined** and decent manner both inside the classroom and in the campus, failing which suitable action may be taken against me as per the rules and regulations of the College and University
6. I will **pay** tuition fees, examination fees and any other **dues** within the stipulated time as required by the University Authorities failing which I will not be permitted to attend the classes and examinations.
7. Violation of any of the undertaking given above will result to my **parents** meeting the concerned HOD/Principal.

Signature of Student: Date:

ACKNOWLEDGEMENT BY PARENT

I have gone through carefully the terms of the above undertaking given by my ward and understand that following these are for his/her own benefits. I also understand that if he/she fails to comply with these terms, will be liable to suitable action as per College/University rules and law. I undertake that he/she will strictly follow the above terms.

Signature of Parent: Date:

Signature of Counsellor Signature of HOD/Principal

Time Table

FY B.TECH DIV 1 CLASS: CL-I ROOM NO: 252

DAY/TIME	MON	TUE	WED	THU	FRI	SAT
09:10 : 10:10	CL143 (B) PCP (628-B) / SC/SS (C) / PY141.01 (A) SDK (273)	CL143 (A) RK (628-B) / SC/SS (B) / PY141.01 (C) USS (273)	ME146 (A/B/C) VP/MO/VHP (726/727)	CL143 (C) RK (628-B) / SC/SS (A) / PY141.01 (B) USS (273)	CL143(Tutorial) PCP (252)	SC/SS (A/B/C)
10:10 : 11:10	CL143 (B) PCP (628-B) / SC/SS (C) / PY141.01 (A) SDK (273)	CL143 (A) RK (628-B) / SC/SS (B) / PY141.01 (C) USS (273)	ME146 (A/B/C) VP/MO/VHP (726/727)	CL143 (C) RK (628-B) / SC/SS (A) / PY141.01 (B) USS (273)	ME146 VP (252)	SC/SS (A/B/C)
11:10 : 12:10	BREAK					
12:10 : 01:10	MA143(Tutorial) RVS/PRM (252/253)	MA143 PRM (252)	MA143 RVS (252)	CL143 NHR (252)	ME146 VP (252)	SC/SS (A/B/C)
01:10 : 02:10	CL143 PCP (252)	CL143 NHR (252)	PY141.01 USS (252)	MA143 RVS (252)	PY141.01 SDK (252)	SC/SS (A/B/C)
02:10 : 2:20	SHORT BREAK					
02:20 : 03:20	MA143 PRM (252)	ME 142 (A/B/C) HB/AV/NG (126)	ME146(Tutorial) VP (252)	Extra lecture/Tutorial/library (A/B/C)	SC/SS (A/B/C)	SC/SS (A/B/C)
03:20 : 04:20	PY141.01 MNS (252)	ME 142 (A/B/C) HB/AV/NG (126)	SC/SS (A/B/C)	Extra lecture/Tutorial/library (A/B/C)	SC/SS (A/B/C)	SC/SS (A/B/C)

MA143	ENGINEERING MATHEMATICS-I	RVS	RAJESH SAVALIA	PRM	PRAGNESH MAKWANA		
CL143	ENGINEERING MECHANICS	NHR	NEHA RAJPUT	PCP	PINAL PATEL	RK	RENU KOSHY
ME146	ENGINEERING GRAPHICS	VP	VIRAL PANARA	MO	MADHAV OZA	VHP	VIKAS PANCHAL
PY141.01	ENGINEERING PHYSICS	MNS	DR. MANAN N. SHAH	SDK	DR. SANNI D. KAPATEL	USS	URVESH S. SONI
ME142	WORKSHOP PRACTICES	HB	HARMISH BHATT	AV	AKASH VYAS	NG	NIPUN GOSAI
SC/SS	ASSIGNMENT PRACTICES/STUDUENT COUNSELLING /REMEDIAL CLASSES						
HSXXX	A COURSE FROM LIBERAL ARTS						

FY B.TECH DIV 2 CLASS: CL-II ROOM NO: 253

DAY/TIME	MON	TUE	WED	THU	FRI	SAT
09:10 : 10:10	PY141.01 USS (253)	CL143 PCP (253)	PY141.01 SDK (253)	CL143 NHR (253)	CL143 (B) MK (628-B) SC/SS (C) PY141.01 (A) MNS (273)	SC/SS (A/B/C)
10:10 : 11:10	PY141.01 MNS (253)	MA143 PKP (253)	ME146 ZS (253)	MA143 RRS (253)		
11:10 : 12:10	BREAK					
12:10 : 01:10	SC/SS (A/B/C)	ME146 ZS (253)	CL143(Tutorial) NHR (253)	ME146 (A/B/C) NG/VP/VHP (726/727)	CL143 PCP (253)	SC/SS (A/B/C)
01:10 : 02:10		ME146(Tutorial) ZS (253)	MA143 RRS (253)		MA143 PKP (253)	
02:10 : 2:20	SHORT BREAK					
02:20 : 03:20	CL143 (C) AJW (628-B) SC/SS (A) PY141.01 (B) SDK (273)	Extra lecture/Tutorial/library (A/B/C)	CL143 (A) MJD (628-B) SC/SS (B) PY141.01 (C) SDK (273)	MA143(Tutorial) RRS/PKP (252/253)	ME 142 (A/B/C) HB//AV/KR (126)	SC/SS (A/B/C)
03:20 : 04:20				SC/SS (A/B/C)		

MA143	ENGINEERING MATHEMATICS-I						
CL143	ENGINEERING MECHANICS	NHR	NEHA RAJPUT	PCP	PINAL PATEL	MK	MEHUL KATAKIYA
ME146	ENGINEERING GRAPHICS	ZS	ZANKHAN SONARA	RP	RUGNESH PATEL	VHP	VIKAS PANCHAL
PY141.01	ENGINEERING PHYSICS	MNS	DR. MANAN N. SHAH	SDK	DR. SANNI D. KAPATEL	USS	URVESH S. SONI
ME142	WORKSHOP PRACTICES	HB	HARMISH BHATT	AV	AKASH VYAS	KR	KAWALJIT RANDHAWA
SC/SS	ASSIGNMENT PRACTICES/STUDUENT COUNSELLING /REMEDIAL CLASSES						
HSXXX	A COURSE FROM LIBERAL ARTS						

FY B.TECH DIV 3 CLASS: ME-I ROOM NO: 255

DAY/TIME	MON	TUE	WED	THU	FRI	SAT
09:10 : 10:10	ME146 (A/B/C) SC/VHP/ZS (726/727)	ME146 SC (255)	CL143 (B) SCP (628-B) | SC/SS (C) | PY141.01 (A) MNS (273)	CL143 PMS (255)	ME146 SC (255)	SC/SS (A/B/C)
10:10 : 11:10		MA143 PRM (255)		MA143 RVS (255)	PY141.01 SDK (255)	
11:10 : 12:10	BREAK					
12:10 : 01:10	CL143(Tutorial) PMS (255)	SC/SS (A/B/C)	SC/SS (A/B/C)	CL143 (A) MK (628-B) | SC/SS (B) | PY141.01 (C) USS (273)	MA143 RVS (255)	CL143 (C) RK (628-B) | SC/SS (A) | PY141.01 (B) SDK (273)
01:10 : 02:10	MA143 PRM (255)				CL143 JKB (255)	
02:10 : 2:20	SHORT BREAK					
02:20 : 03:20	PY141.01 MNS (255)	PY141.01 USS (255)	ME 142 (A/B/C) HB/AV/RP (126)	HS 101A/(Extra lectures/Tutorial/library) (A/B/C)	MA143 (Tutorial) RVS/PRM (255/256)	CL143 (C) RK (628-A) | SC/SS (A) | PY141.01 (B) MNS (273)
03:20 : 04:20	ME146(Tutorial) SC (255)	CL143 JKB (255)			SC/SS (A/B/C)	

MA143	ENGINEERING MATHEMATICS-I	RVS	RAJESH SAVALIA	PRM	PRAGNESH MAKWANA		
CL143	ENGINEERING MECHANICS	JKB	JAY BHAVSAR	PMS	PINKI SHARMA	SCP	SARASWATI PATHARIYA
ME146	ENGINEERING GRAPHICS	SC	SAGAR CHOKSHI	VHP	VIKAS PANCHAL	RP	RUGNESH PATEL
PY141.01	ENGINEERING PHYSICS	MNS	DR. MANAN N. SHAH	SDK	DR. SANNI D. KAPATEL	USS	URVESH S. SONI
ME142	WORKSHOP PRACTICES	HB	HARMISH BHATT	AV	AKASH VYAS	RP	RUGNESH PATEL
SC/SS	ASSIGNMENT PRACTICES/STUDUENT COUNSELLING /REMEDIAL CLASSES						
HSXXX	A COURSE FROM LIBERAL ARTS						

FY B.TECH DIV 4 CLASS: ME-II ROOM NO: 256

<table>
<tr><th>DAY/TIME</th><th>MON</th><th colspan="3">TUE</th><th>WED</th><th colspan="3">THU</th><th>FRI</th><th colspan="3">SAT</th></tr>
<tr><td>09:10 : 10:10</td><td>MA143
PRM
(256)</td><td colspan="3">ME146
VHP
(256)</td><td>PY141.01
USS
(256)</td><td colspan="3">MA143(Tutorial)
RVS/PKP
(256/252)</td><td rowspan="2">SC/SS
(A/B/C)</td><td rowspan="2">CL143
(C)
MJD
(628-A)</td><td rowspan="2">SC/SS
(A)</td><td rowspan="2">PY141.01
(B)
SDK
(273)</td></tr>
<tr><td>10:10 : 11:10</td><td>CL143
PMS
(256)</td><td colspan="3">CL143
PMS
(256)</td><td>SC/SS
(A/B/C)</td><td colspan="3">PY141.01
SDK
(256)</td></tr>
<tr><td>11:10 : 12:10</td><td colspan="12">BREAK</td></tr>
<tr><td>12:10 : 01:10</td><td>ME146(Tutorial)
VHP
(256)</td><td colspan="3">PY141.01
MNS
(256)</td><td rowspan="2">ME146
(A/B/C)
VHP//SC/NG
(726/727)</td><td colspan="3">CL143(Tutorial)
JKB
(256)</td><td>ME146
VHP
(256)</td><td colspan="3" rowspan="2">SC/SS
(A/B/C)</td></tr>
<tr><td>01:10 : 02:10</td><td>SC/SS
(A/B/C)</td><td colspan="3">MA143
RVS
(256)</td><td colspan="3">MA143
PRM
(256)</td><td>MA143
RVS
(256)</td></tr>
<tr><td>02:10 : 2:20</td><td colspan="12">SHORT BREAK</td></tr>
<tr><td>02:20 : 03:20</td><td rowspan="2">ME 142
(A/B/C)
VM/NG/HB
(126)</td><td rowspan="2">CL143
(B)
MK
(628-B)</td><td rowspan="2">SC/SS
(C)</td><td rowspan="2">PY141.01
(A)
MNS
(273)</td><td rowspan="2">Extra lecture/Tutorial/library
(A/B/C)</td><td rowspan="2">CL143
(A)
HKS
(628-B)</td><td rowspan="2">SC/SS
(B)</td><td rowspan="2">PY141.01
(C)
USS
(273)</td><td>SC/SS
(A/B/C)</td><td rowspan="2">CL143
(C)
MJD
(628-B)</td><td rowspan="2">SC/SS
(A)</td><td rowspan="2">PY141.01
(B)
SDK
(273)</td></tr>
<tr><td>03:20 : 04:20</td><td>CL143
JKB
(256)</td></tr>
</table>

MA143	ENGINEERING MATHEMATICS-I	**RVS**	RAJESH SAVALIA	**PRM**	PRAGNESH MAKWANA	**PKP**	PURVAK PATEL
CL143	ENGINEERING MECHANICS	**JKB**	JAY BHAVSAR	**PMS**	PINKI SHARMA	**HKS**	DR. HITESHRI SHASTRI
ME146	ENGINEERING GRAPHICS	**VHP**	VIKAS PANCHAL	**SC**	SAGAR CHOKSHI	**NG**	NIPUN GOSAI
PY141.01	ENGINEERING PHYSICS	**MNS**	DR. MANAN N. SHAH	**SDK**	DR. SANNI D. KAPATEL	**USS**	URVESH S. SONI
ME142	WORKSHOP PRACTICES	**VM**	VISHAL MEHTA	**NG**	NIPUN GOSAI	**HB**	HARMISH BHATT
SC/SS	ASSIGNMENT PRACTICES/STUDUENT COUNSELLING /REMEDIAL CLASSES						
HSXXX	A COURSE FROM LIBERAL ARTS						

FY B.TECH DIV 5 CLASS: EE-I ROOM NO: 252/253/255/256

<table>
<tr><th>DAY/TIM</th><th>MON</th><th colspan="3">TUE</th><th>WED</th><th>THU</th><th colspan="3">FRI</th><th>SAT</th></tr>
<tr><td>09:10 : 10:10</td><td>MA143
RVS
(252)</td><td colspan="3">MA143
PRM
(252)</td><td>ME146
SC
(255)</td><td>SC/SS
(A/B/C)</td><td colspan="3">CL143
MJD
(253)</td><td rowspan="2">SC/SS
(A/B/C)</td></tr>
<tr><td>10:10 : 11:10</td><td>PY141.01
USS
(252)</td><td colspan="3">PY141.01
MNS
(252)</td><td>PY141.01
SDK
(255)</td><td>CL143(Tutorial)
MK
(252)</td><td colspan="3">ME146(Tutorial)
SC
(253)</td></tr>
<tr><td>11:10 : 12:10</td><td colspan="10">BREAK</td></tr>
<tr><td>12:10 : 01:10</td><td rowspan="2">ME 142
(A/B/C)
HB/AV/VM
(126)</td><td rowspan="2">CL143
(B)
HKS
(628-B)</td><td rowspan="2">SC/SS
(C)</td><td rowspan="2">PY141.01
(A)
USS
(273)</td><td rowspan="2">Extra lecture/Tutorial/library
(A/B/C)</td><td>MA143
PRM
(253)</td><td rowspan="2">CL143
(A)
PCP
(628-B)</td><td rowspan="2">SC/SS
(B)</td><td rowspan="2">PY141.01
(C)
MNS
(273)</td><td rowspan="2">SC/SS
(A/B/C)</td></tr>
<tr><td>01:10 : 02:10</td><td>ME146
SC
(253)</td></tr>
<tr><td>02:10 : 2:20</td><td colspan="10">SHORT BREAK</td></tr>
<tr><td>02:20 : 03:20</td><td>CL143
MJD
(253)</td><td colspan="3" rowspan="2">ME146
(A/B/C)
SC/VP/RP
(726/727)</td><td>MA143
RVS
(252)</td><td rowspan="2">SC/SS
(A/B/C)</td><td rowspan="2">CL143
(C)
PCP
(628-B)</td><td rowspan="2">SC/SS
(A)</td><td rowspan="2">PY141.01
(B)
USS
(273)</td><td rowspan="2">SC/SS
(A/B/C)</td></tr>
<tr><td>03:20 : 04:20</td><td>MA143(Tutorial)
RVS/PKP
(253/255)</td><td>CL143
MK
(252)</td></tr>
</table>

MA143	ENGINEERING MATHEMATICS-I	RVS	RAJESH SAVALIA	PRM	PRAGNESH MAKWANA	PKP	PURVAK PATEL
CL143	ENGINEERING MECHANICS	MJD	MEGHA DESAI	MK	MEHUL KATAKIYA	HKS	DR. HITESHRI SHASTRI
ME146	ENGINEERING GRAPHICS	SC	SAGAR CHOKSHI	VP	VIRAL PANARA	RP	RUGNESH PATEL
PY141.01	ENGINEERING PHYSICS	MNS	DR. MANAN N. SHAH	SDK	DR. SANNI D. KAPATEL	USS	URVESH S. SONI
ME142	WORKSHOP PRACTICES	HB	HARMISH BHATT	AV	AKASH VYAS	VM	VISHAL MEHTA
SC/SS	ASSIGNMENT PRACTICES/STUDUENT COUNSELLING /REMEDIAL CLASSES						
HSXXX	A COURSE FROM LIBERAL ARTS						